

Access, Capacity, Learning

Complete discussion notebook ·

Help useful work reach its next step.

Three priorities for the next five years: widen access, build capacity, and make learning easier to use.

These are proposals drawn from my work in independent research, open-source software, and community partnerships.

Choose a priority above, then a topic to bring into the discussion.

A point to bring to the room

Support should connect people to a useful next step—and leave enough evidence to show what that support changed.

From discussion to a practical pilot

The convening asks us to name barriers, understand their causes, identify an intervention and its partners, and define progress. The three priorities above follow that structure.

Access focuses on supported referrals. Capacity focuses on engineering and validation. Learning focuses on reusable evidence and better decisions about continued support.

These are participant proposals for the National Innovation Roadmap. Track tags indicate where a topic may fit; they are editorial suggestions.

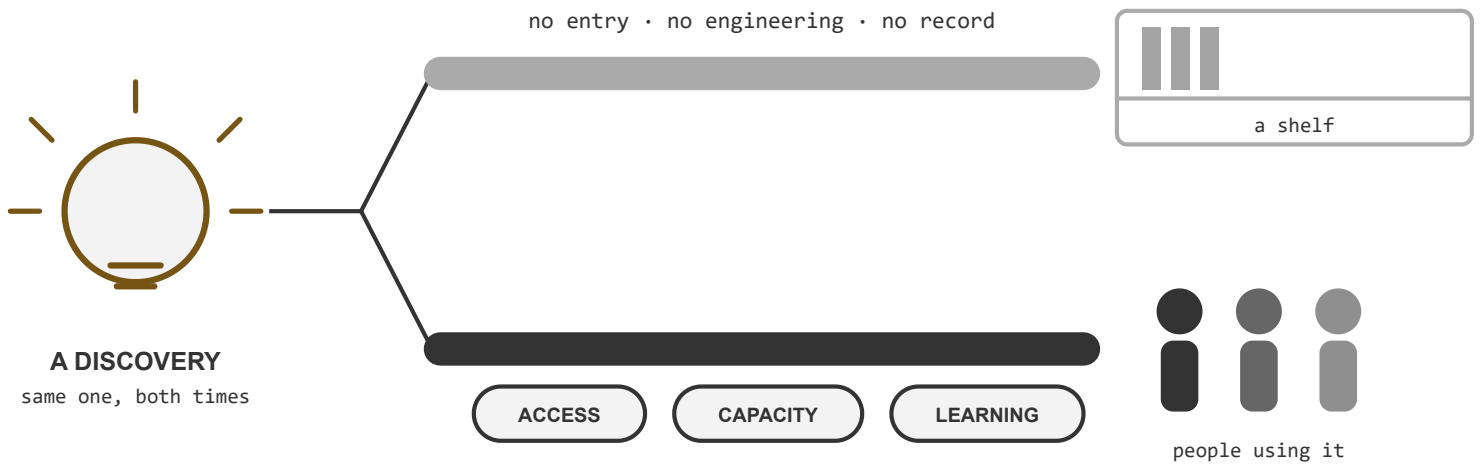
Label	Meaning
documented	A result or approach described in retained source records; not independently reproduced for this presentation.
in development	Implementation work with remaining qualification or readiness questions.
proposed	A recommendation or possible pilot; not a claim that it already exists.

Sources: [Widener convening](#) · [Convening agenda](#)

Agenda connection: [reflection questions, pp. 12–15](#); [breakout prompts and tracks, pp. 5–6](#).

Two ways to see the idea

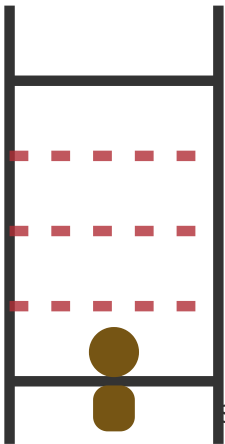
Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



The discovery is identical on both tracks. Only the support underneath it differs.

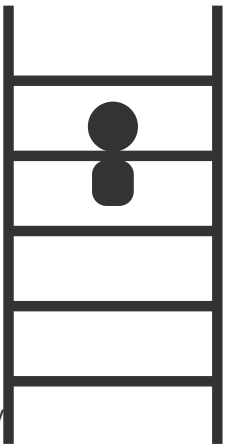
Same discovery. Two endings. The difference is underneath.

the goal is up here



rungs missing

VS



access · capacity · learning

Same climber, same height. The rungs are the policy.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Independent researchers

Who has useful work but cannot reach the people, infrastructure, or support needed to advance it?

My example

Good work starts in places without a grants office: a home lab, an open-source project, a small nonprofit. Being eligible for a program is not the same as having a usable path into it.

Question for the room. Who owns the first handoff when someone has promising work but no institution behind them?

A practical next step

Pilot a supported referral path for independent researchers and small teams. Record where the handoff breaks down.

Progress would mean: Time to a useful connection; completed handoffs; reasons people are turned away.

Who needs to be involved

Program navigators, independent researchers, research institutions, and regional partners.

Research stage: Discovery → Translation.

More context

- Useful work can start outside universities, established companies, and federal laboratories.
- Independent contributors usually lack grants staff, technology-transfer offices, institutional infrastructure, and professional introductions.
- Evaluate applicants on relevant evidence, capability, and potential contribution, not affiliation alone.

Evidence and technical detail

Breakout point 2 asks for root causes. A barrier to investigate is the assumption that an institution supplies staff, infrastructure, and introductions. Its importance will vary by program and applicant.

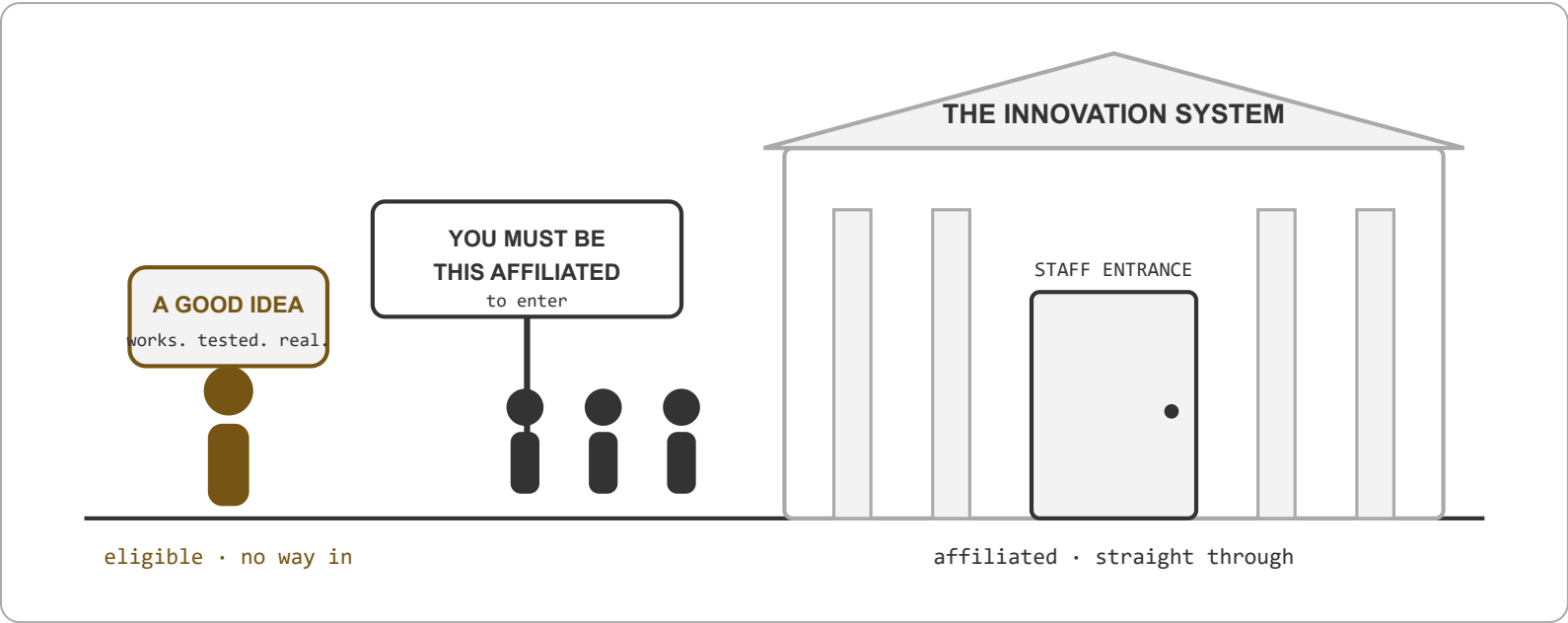
Breakout point	Contribution
1. Challenge	Capable independent contributors cannot convert eligibility into entry.
2. Root cause	Support systems assume institutional affiliation and staff.
3. Existing models	SBIR/STTR for small businesses; OSF and Registered Reports for open research practice.
4. Gap	Identify where individuals, informal teams, and small nonprofits lack an accountable handoff.
5. Recommendation	Supported entry pathways with a named navigator.
6. Stakeholders	NSF TIP, SBIR/STTR, EPSCoR institutions, open-source foundations, community organizations.

Breakout point	Contribution
7. Measures	Completed connections and progress toward validation, not portal visits.

Agenda connection: [Resource Guide, p. 14](#). Suggested breakout: Track C · Innovation infrastructure. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

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Eligible is not the same as admitted.

APPLICATION

Name

filled in

Project

filled in

Institution *

* required – cannot continue

One field, and the whole form is unreachable. That is the barrier, stated exactly.

One required field, and the whole form is unreachable.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Finding support

How can we reduce the time spent finding the right program or partner?

My example

One front door for funding, compute, storage, technical help, research partners, and commercialization support. The important part is not the website. It is the named person or office responsible for what happens after the match.

proposed

A practical next step

Connect existing directories to a named navigator who can make a warm referral and confirm the outcome.

Progress would mean: Referrals that reach a suitable partner, rather than portal visits alone.

Who needs to be involved

Program staff, technical assistance providers, and people seeking support.

Research stage: Across the research journey.

More context

- Profiles for individuals, informal open-source teams, nonprofits, and small businesses.
- Opportunity records that state eligibility, affiliation rules, resources offered, and when those details were last verified.
- Matching by the contributor's actual need: compute, storage, validation, research partner, maintenance funding, or commercialization support.
- A named navigator or receiving organization responsible for following through.
- Tracking whether a referral produced a useful connection, and why unsuccessful referrals stopped.

Evidence and technical detail

Funding search already exists, for example [Simpler.Grants.gov](#). The proposal adds what search cannot do: responsibility for the handoff. AI can help discover and explain matches. Program owners stay responsible for eligibility decisions and ambiguous cases.

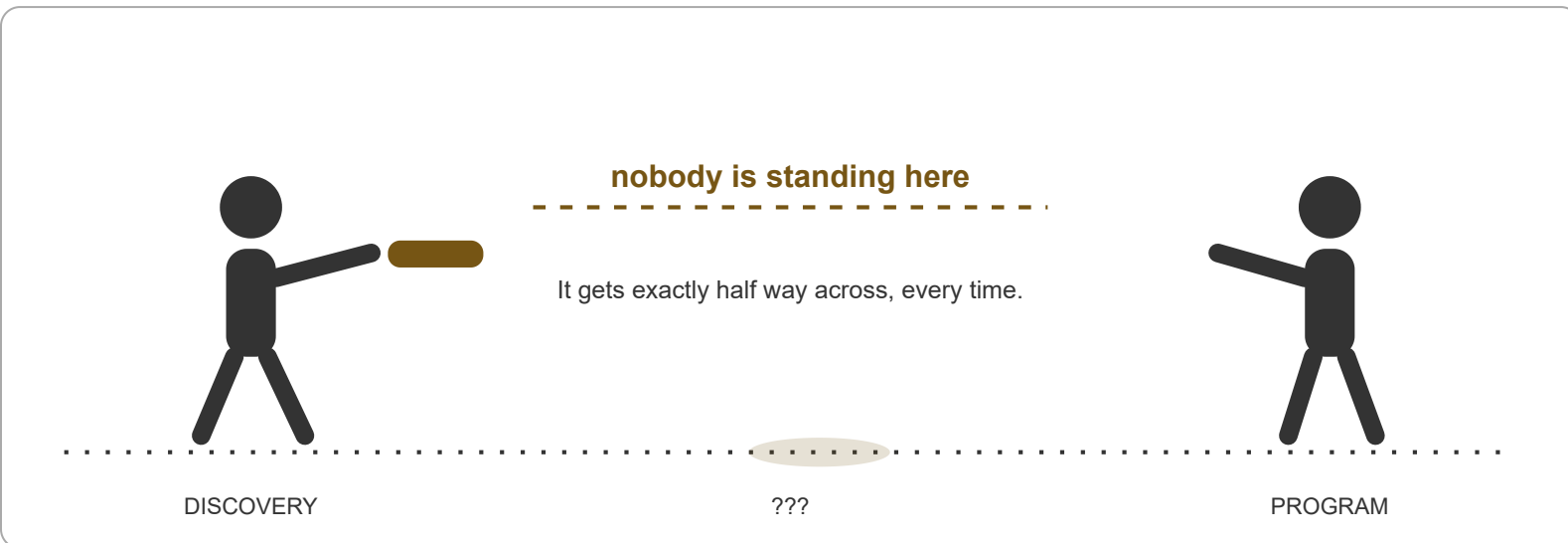
Portal function	Who does it	Why
Find and explain candidate matches	AI assistance	Fast, broad, cheap to repeat
Eligibility determination	Program owner	Accountable, appealable
Ambiguous or consequential cases	Human navigator	Judgment and follow-through
Referral outcome tracking	Portal record	Shows where applicants get stuck

A concrete use case appears later on this site: a nonprofit with a community and worthwhile questions, but no research partner.

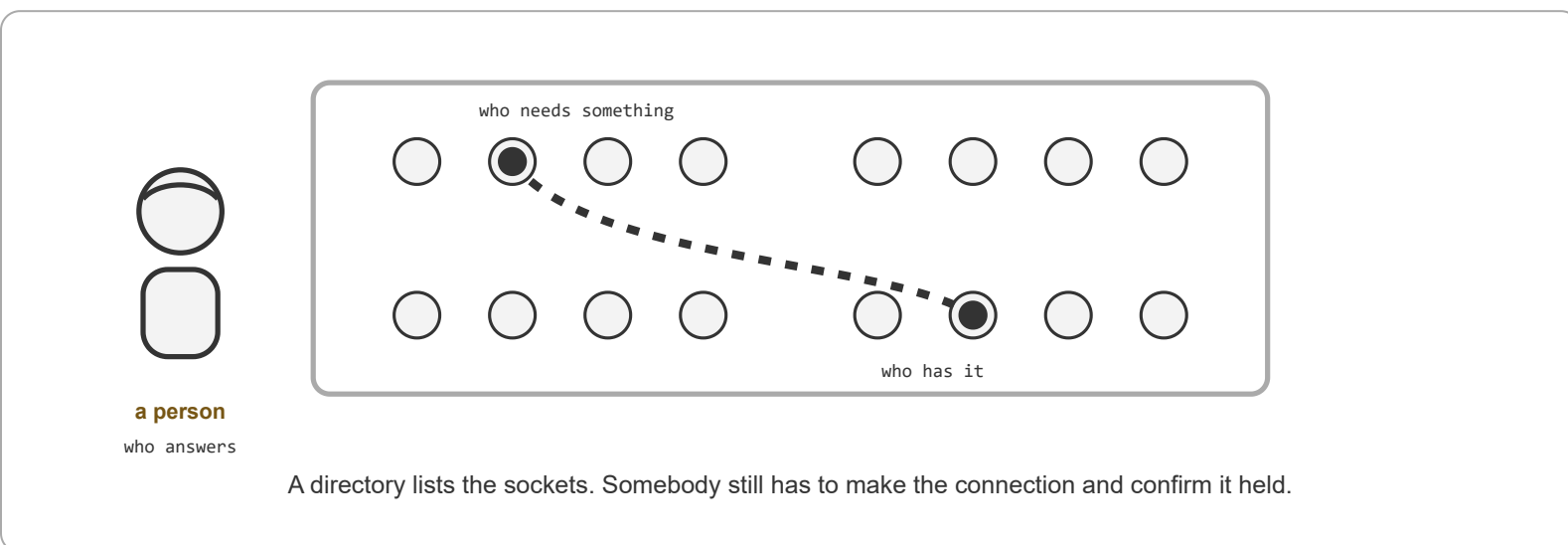
Agenda connection: [Resource Guide, p. 12](#). Suggested breakout: Track B · Ecosystem best practices. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

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It gets exactly half way across, every time.



A directory lists the sockets. Somebody still has to make the connection.

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Open-source stewardship

What support keeps a research tool usable after its first release?

My example

Research runs on software that someone has to test, document, secure, maintain, and support. That work is what lets a discovery be used by other researchers, nonprofits, businesses, and public agencies.

A practical next step

Treat maintenance, documentation, testing, and user support as explicit work in a bounded pilot.

Progress would mean: External reuse, maintenance continuity, and reproducible results.

Who needs to be involved

Maintainers, research software engineers, users, and funding partners.

Research stage: Translation → Adoption.

More context

- Maintenance and adoption work deserve recognition alongside new inventions.
- Support should reach the people maintaining critical dependencies, including independent contributors.
- Reuse lowers adoption costs for everyone downstream.

Evidence and technical detail

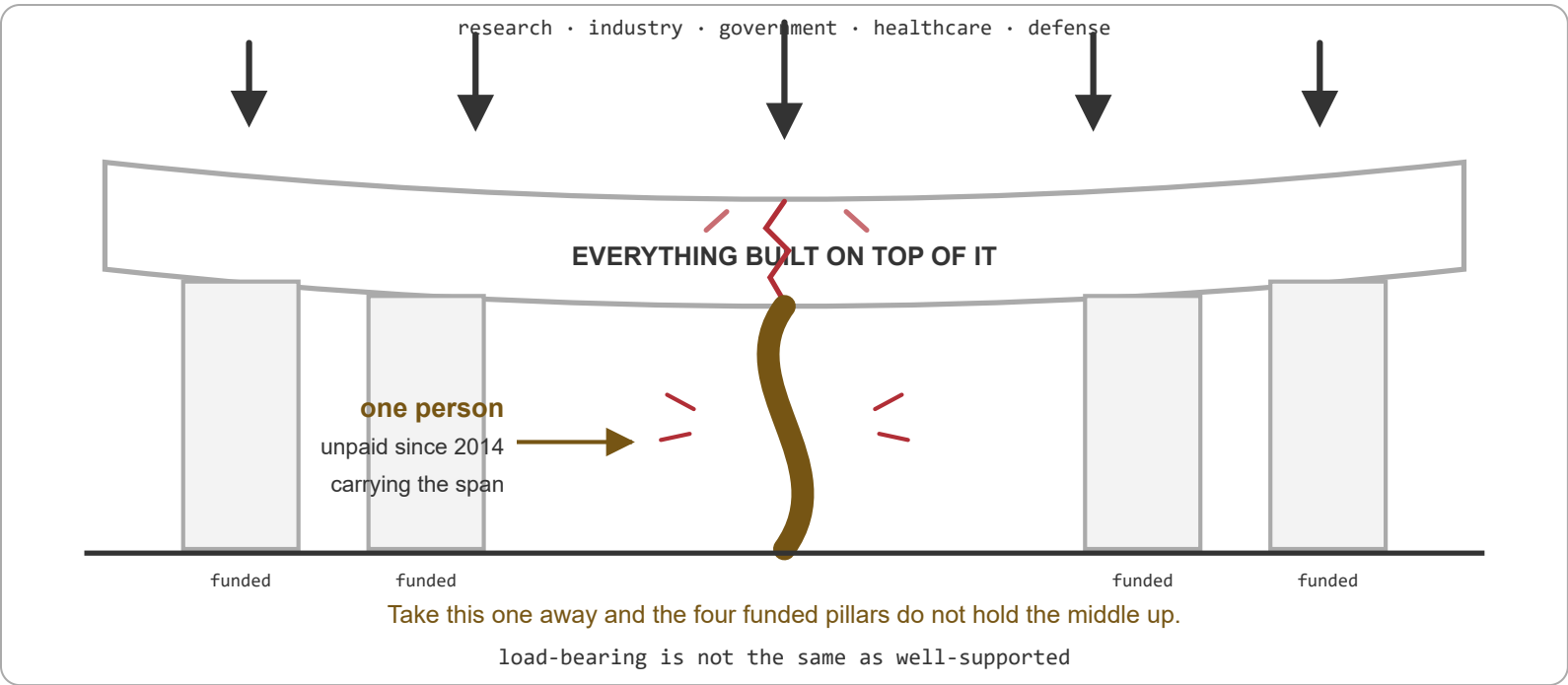
Open-source projects provide reusable research tools. Maintaining them can require testing, documentation, security work, and support that a research grant does not always cover. The proposal is to recognize and fund that work explicitly.

The Mojo ports and Rust/Mojo development attempts in this portfolio offer cases for discussion. They do not establish that every rewrite improves performance or that every open-source project lacks funding.

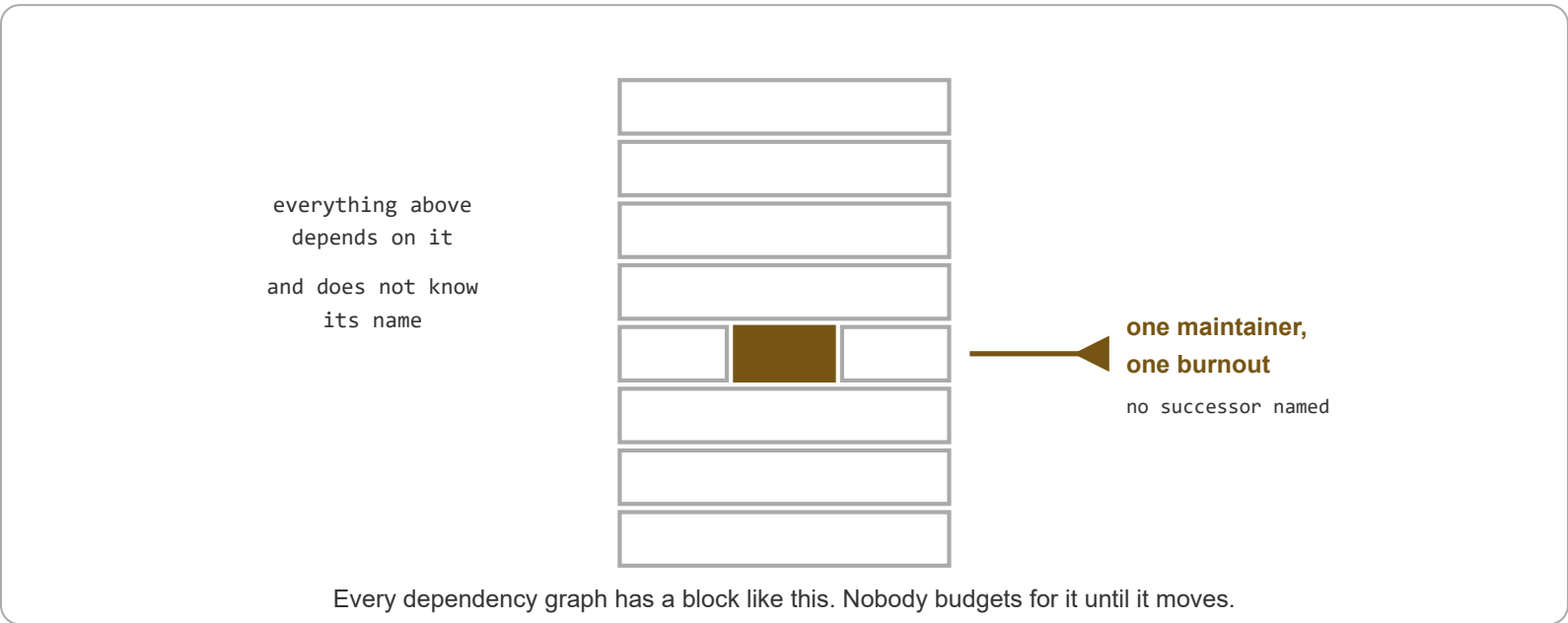
Agenda connection: [Resource Guide, p. 2](#). Suggested breakout: Track C · Innovation infrastructure. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Load-bearing is not the same as well-supported.



Every dependency graph has a block like this.

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Software qualification

What evidence is needed before a promising tool is ready for the next stage?

My example

We rewrote part of our scientific computing tools in Mojo and checked the new version against the established one. Across 24 test cases the two programs gave exactly the same answers. That was only the first test.

We then compared the answers with independent mathematical references. The higher-precision version passed. The lower-precision version failed.

Two programs can agree perfectly and still miss the target. Agreement and accuracy are different questions.

documented

Scope: S02-I, 20 August 2026. Overall protocol: failed. Applies to the tested stencil and environment. No new GPU run, speed qualification, or physical-feasibility finding.

Sources: Pinned qualification report

A practical next step

Use a small qualification case with fixed criteria. Keep agreement, numerical accuracy, and performance as separate claims.

Progress would mean: A reviewer can identify the tested scope, failed criteria, and evidence needed for the next decision.

Who needs to be involved

Domain researchers, numerical specialists, software engineers, and prospective users.

Research stage: Translation → Validation.

More context

Think of two calculators showing the same answer. That is encouraging: they are consistent. But if both round away a small number that matters, both give the same wrong answer. Agreement is a useful check. Accuracy needs an independent reference.

Check	Result	Plain meaning
Same input bytes read by both programs	Pass, 24 of 24	Both started from the same data
Same output bytes	Pass, 24 of 24	No disagreement between HIP and Mojo
Float64 against independent reference	Pass	Higher precision met the pre-set limits
Float32 against independent reference	Fail	Lower precision exceeded the limits
Overall protocol	Fail	The full requirement set was not met

Check	Result	Plain meaning
Speed	Not qualified here	No performance qualification is established by these results

- Modernization should fix a shown limitation in correctness, maintainability, portability, security, or performance. Language choice alone proves nothing.
- The accuracy limits were set before the run, so any later change to those limits requires a new protocol version.
- The failed result is kept and published. It records which tested criteria passed and which need further work.

Evidence and technical detail

The tested operation is a *stencil*: it compares a grid point with its neighbors to estimate how a quantity changes across space. Such calculations subtract nearly equal numbers and then magnify what remains, so rounding matters even in a single evaluation. Float32 keeps about seven significant digits; Float64 keeps about 15 to 16.

Campaign S02-I covered two precisions, three grid sizes (64^3 , 128^3 , 256^3), and four input families (quadratic, normalized cubic, asymmetric, sinusoidal), with analytic and independent CPU references.

Precision	Max relative L2 error	Frozen limit	Max scaled L^∞ error	Frozen limit
Float64	5.98×10^{-12}	1×10^{-11}	5.83×10^{-11}	1×10^{-10}
Float32	2.32488×10^{-3}	1×10^{-4}	1.5625×10^{-2}	1×10^{-3}

Float32 exceeded its limits by roughly 23 and 16 times. The repository attributes the shared failure to finite-precision conditioning and proposes a dedicated study of cancellation, scaling, and resolution.

Options on the table

Option	Benefit	Required check
Use Float64 now for the qualified stencil	Already meets the accuracy gates	Twice the memory per value; real speed cost depends on hardware
Reformulate the calculation	May preserve small differences without extra precision everywhere	Algebraically equal expressions can behave differently on a computer
Mixed precision	Could cut memory and compute cost	Cannot restore detail already lost in Float32 inputs
Stronger independent references	Locates the error source	The reference itself needs validation
Benchmark after accuracy passes	Evidence-based cost decisions	Hardware changes do not fix low-precision arithmetic

Three separate questions stay separate: do HIP and Mojo agree; do their answers meet an independent reference; does the model agree with physical evidence? This campaign answers the first two for one kernel on one system. It says nothing about physical feasibility.

The wider Rust and Mojo portfolio

Repository	What is there	Honest description
KosMojAMD	Six GPU kernel ports (Laplacian, diffusion, derivative stencil, RK4, atomics, STREAM Triad), AMD gfx1103 evidence, HIP comparisons, the 24-case assessment	documented kernel modernization with bounded hardware validation
Computational-Spacetime-Dynamics-Research	Comparison reports, run logs, numerical checks, blocked attempts, corrected findings	documented research campaign that produced reusable tooling
LLM_Engine_Rust-Over-GGML	Archived Rust serving layer, GGML and Mojo backends, controlled 16-arm comparison, unfinished wave 2	documented bounded comparison with retained limitations
LLM_Engine_MojoRust	Gated Rust/Mojo engine rebuild; accelerator and serving milestones incomplete	in development
GKOS-Engine-Rust PR 18	Retained executions, differential cases, open qualification gaps	in development

Sources: [Assessment 2026-08-20](#) · [S02-I disposition](#) · [Preregistered protocol](#) · [Floating point background \(NVIDIA\)](#)

Evidence boundary: inspected 2026-09-17 at commit 0477d6e. GPU tests not rerun; the externally retained 1.2 GiB raw record was not inspected.

Relative L2 and scaled L^∞ are separate protocol metrics, not universal pointwise percentage errors. The referenced raw dataset was not independently inspected for this presentation.

Agenda connection: [Resource Guide](#), p. 5. Suggested breakout: Track A · Barriers to translation. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.

Two programs can agree and still miss the target.

Mojo / AMD stencil qualification • S02-I • 20 August 2026

TEST 1 · AGREEMENT

Do both programs give the same answer?

HIP ↔ Mojo

24 / 24 cases identical

PASS · compared output bytes

TEST 2 · ACCURACY

Does that answer meet the reference limits?

Float64 · PASS

Float32 · FAIL

Independent mathematical reference

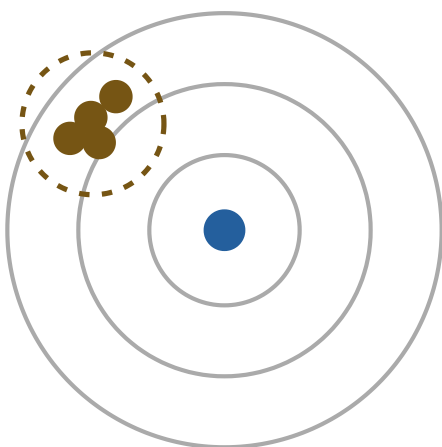
Use qualified Float64. Investigate lower precision separately.

Overall S02-I: FAIL • Findings apply to the tested stencil and environment.

Neither test proves physical feasibility. This campaign did not qualify speed.

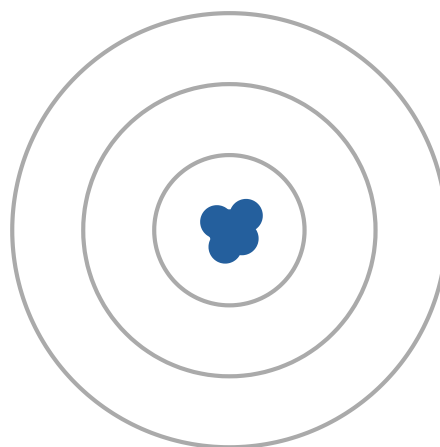
Agreement and accuracy are separate tests.

HIP and Mojo, Float32



Both pairs agree with each other. Only one pair agrees with the target.
TIGHT GROUP · WRONG PLACE

HIP and Mojo, Float64



TIGHT GROUP · ON TARGET

Both pairs agree with each other. Only one agrees with the target.

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Evidence and decisions

How can a partnership show why it made a decision, and revise it when evidence changes?

My example

When AI helps with research, someone later has to ask: what evidence supports this claim, what is still uncertain, and what was this action allowed to do? GKOS is my developing approach to keeping those links.

Provenance lets people inspect a claim. It does not make the claim true.

in development

A practical next step

Test one evidence record that connects a claim, its sources, review, limits, and next decision.

Progress would mean: A new reviewer can trace the claim to evidence and explain what changed after review.

Who needs to be involved

Researchers, evidence reviewers, tool builders, and the people using the result.

Research stage: Across the research journey.

More context

- AI-assisted work needs records of sources, transformations, checks, authorization, and resulting actions.
- Reviewers should be able to see what supports a claim, what remains uncertain, and what an action was authorized to do.
- Corrections stay visible as new versions instead of silently rewriting history.

Evidence and technical detail

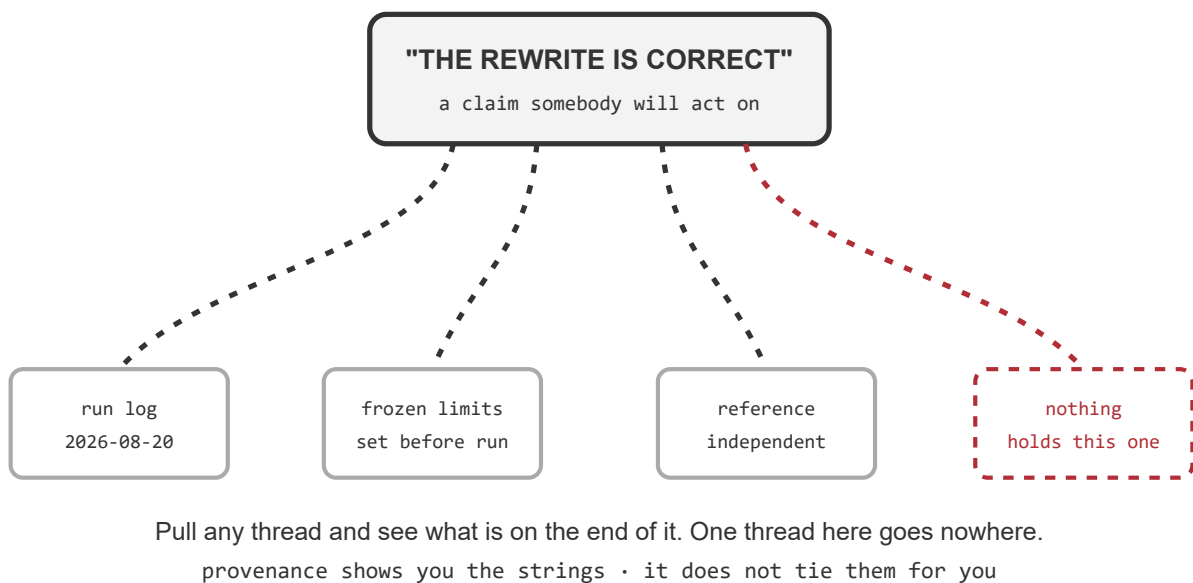
Repository	Role	Status
<u>gkos-standard</u>	Pre-standard for evidence lineage, authority, dispositions, governance responsibilities, conformance fixtures	documented
<u>GKOS-Engine</u>	TypeScript reference engine: parsing, validation, lineage, knowledge graph, retrieval, citations	documented
<u>GKOS-Engine-Rust</u>	Ground-up Rust implementation; implementation and qualification records; consult the current repository for status	in development
<u>GKOS-Engine-Lite</u>	Rust retrieval and restricted-profile work; conformance and adversarial cases	in development
<u>model-capability-manifest-standard</u>	Declare AI capability through structured, testable, versioned evidence rather than model names	proposed

Proposed division of labor: AI extracts metadata and proposes classifications and related records; deterministic checks enforce allowed values, required evidence, and consistency; humans review ambiguity and consequential interpretation. Deterministic rules can enforce record integrity. They cannot make scientific interpretation automatically correct, so conflicting interpretations stay attributable and inspectable.

Agenda connection: [Resource Guide](#), p. 13. Suggested breakout: Track B · Ecosystem best practices. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Provenance shows you the strings. It does not tie them for you.

CHAIN OF CUSTODY		
collected	2026-08-20 · rig gfx1103	signed
measured	frozen limits, set 08-14	signed
compared	independent CPU reference	signed
concluded	"the rewrite is correct"	unsigned

Three steps are accounted for. The step from result to claim is the one nobody signed.

The step from result to claim is the one nobody signed.

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AI and verification

Where can AI reduce friction, and where must human judgment remain?

My example

AI can accelerate code, analysis, documentation, and proposed experiments. Checking that output still takes time and expertise. Confidence and a fluent explanation are not acceptance criteria.

Question for the room. As producing research artifacts becomes cheaper, how do we keep checking them from becoming the next bottleneck?

A practical next step

Pair AI-assisted work with deterministic checks, explicit evidence requirements, and an accountable human reviewer.

Progress would mean: Review time and correction rates, together with the quality of the final evidence.

Who needs to be involved

Researchers, AI tool builders, domain reviewers, and program evaluators.

Research stage: Across the research journey.

More context

- AI capability varies by task. Test it per task instead of trusting a model name.
- Use deterministic checks where requirements are explicit. Use qualified human judgment where interpretation is needed.
- Budget verification capacity alongside acceleration. One without the other produces volume, not knowledge.

Evidence and technical detail

Two working examples from this portfolio show the shape of the answer.

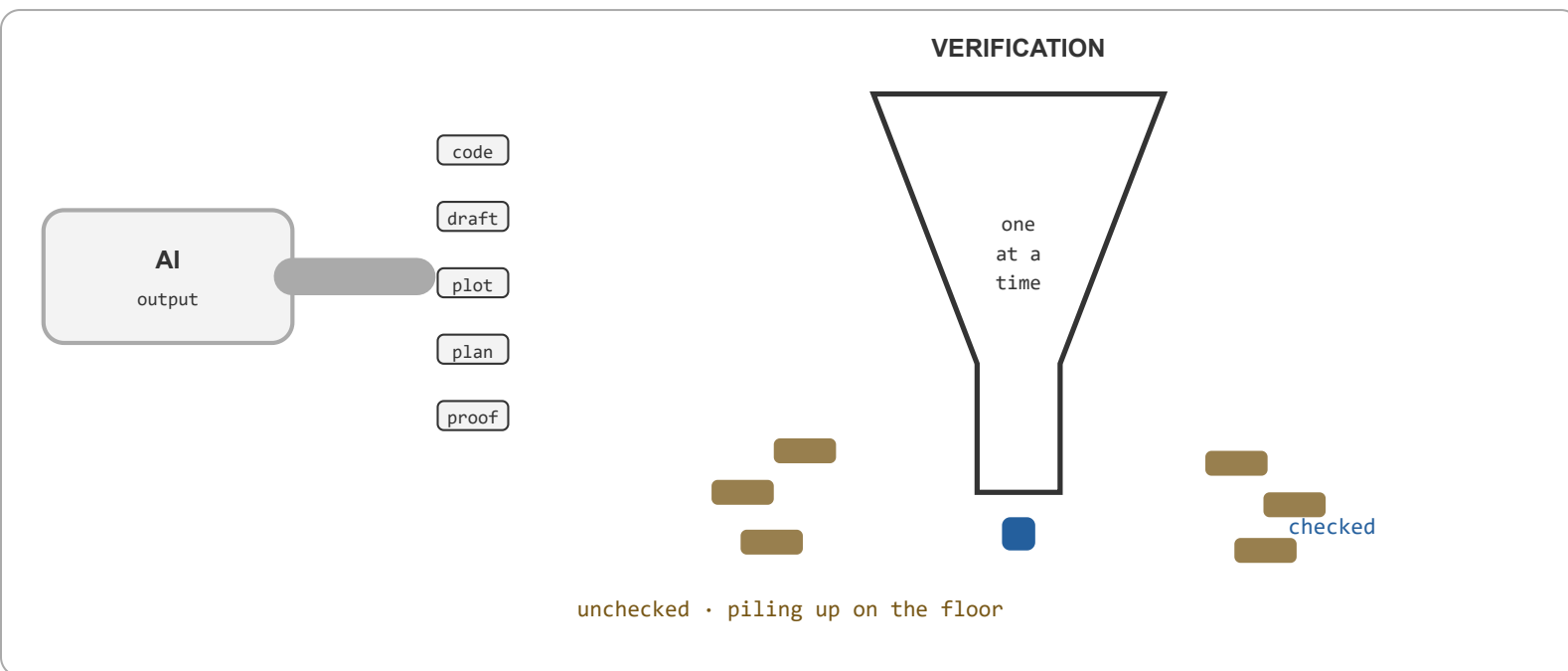
1. **Pre-set gates.** The Mojo accuracy limits were frozen before the run. When Float32 failed, the failure stood. A later scientifically justified change requires a new protocol; the original failure remains on record.
2. **Retained corrections.** An earlier finding of `TOOLCHAIN_ABSENT` was later corrected: an agent had checked global binaries and missed an existing Pixi environment. The record keeps both the wrong diagnosis and the correction, so the next person does not repeat it.

The capability-manifest idea belongs here too: if a model or agent declares what it can do in a structured, testable form, verification can be planned instead of improvised.

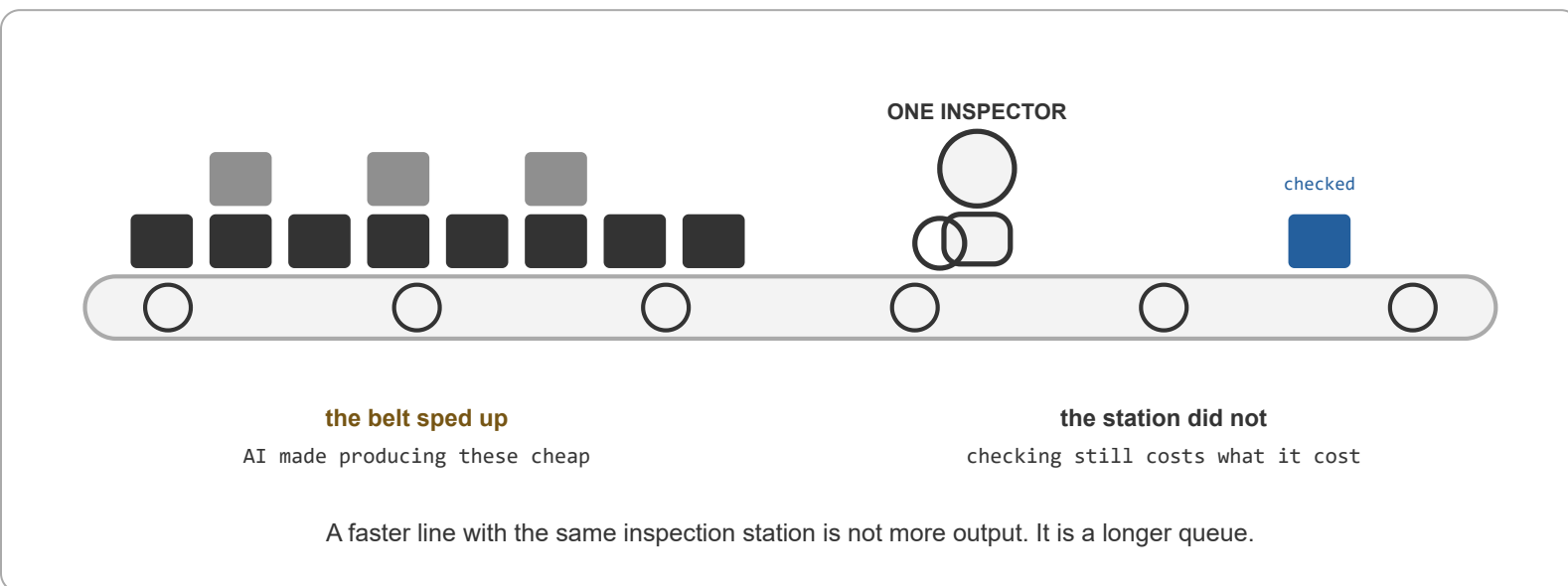
Sources: [Retest report with correction](#)

Two ways to see the idea

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Production got cheap. Checking did not.



A faster line with the same inspection station is a longer queue.

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Connected research records

How can existing resources connect and scale without losing their custodians or context?

My example

A Unified Research Library could connect existing repositories through a shared catalog. Original custodians would keep their data, permissions, and expertise.

Make it easier to find a question, its evidence, and what was learned across projects.

proposed

A practical next step

Start with a small shared catalog that points to original records and preserves permissions, attribution, corrections, and disagreements.

Progress would mean: Time to find relevant evidence; usable cross-references; fewer missing source records.

Who needs to be involved

Repository custodians, librarians, researchers, and standards partners.

Research stage: Discovery → Reuse.

More context

- Link protocols, attempts, code, environments, results, reviews, and later decisions.
- Preserve custodianship, attribution, licensing, and access restrictions.
- Record why an attempt stopped and what would be needed to continue.
- Keep negative results that resolve uncertainty, alongside implementation and infrastructure failures.
- Do not treat a missing record as proof that an experiment never ran. Use *unknown*.

Evidence and technical detail

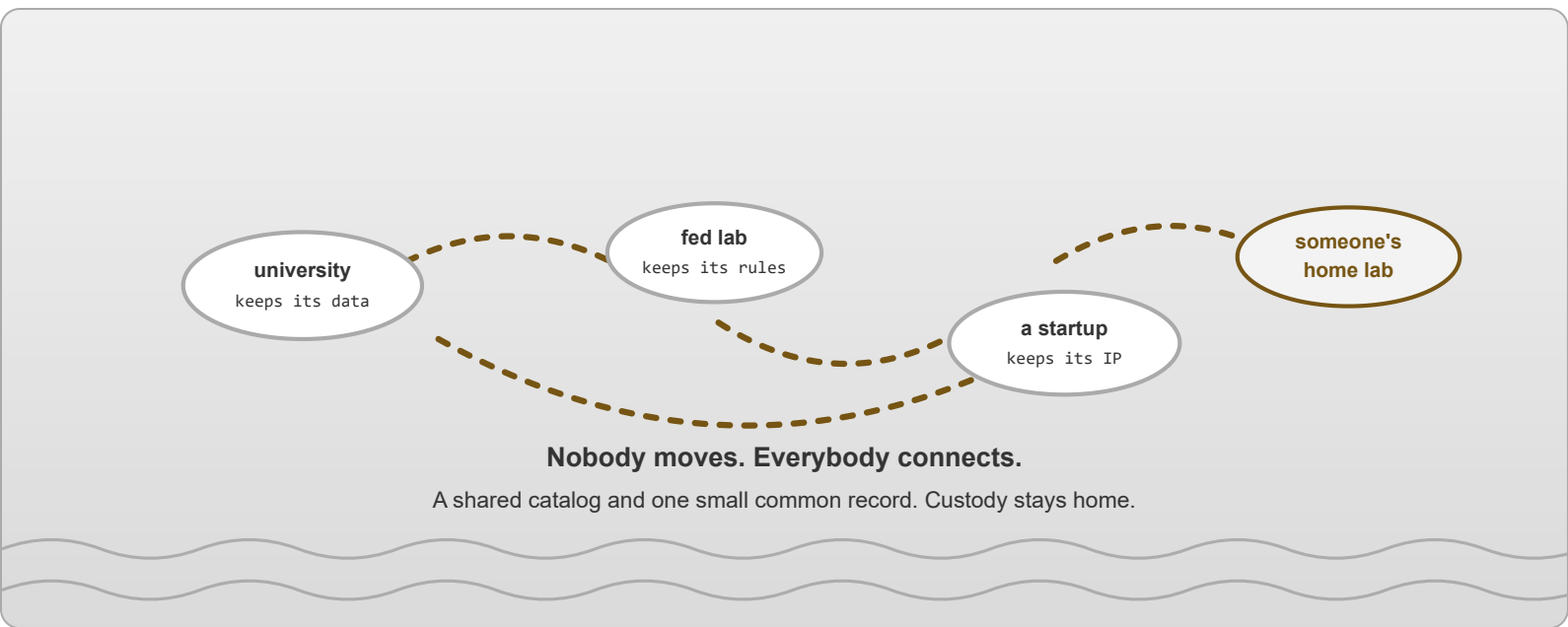
Start with interoperable records, stable source identifiers, and a small shared vocabulary. Preserve licensing, attribution, access restrictions, corrections, and disagreements. AI could propose categories and related records; deterministic checks can enforce structure, while people review ambiguous interpretations.

Records reconstructed later must be labeled as reconstruction, not preregistration.

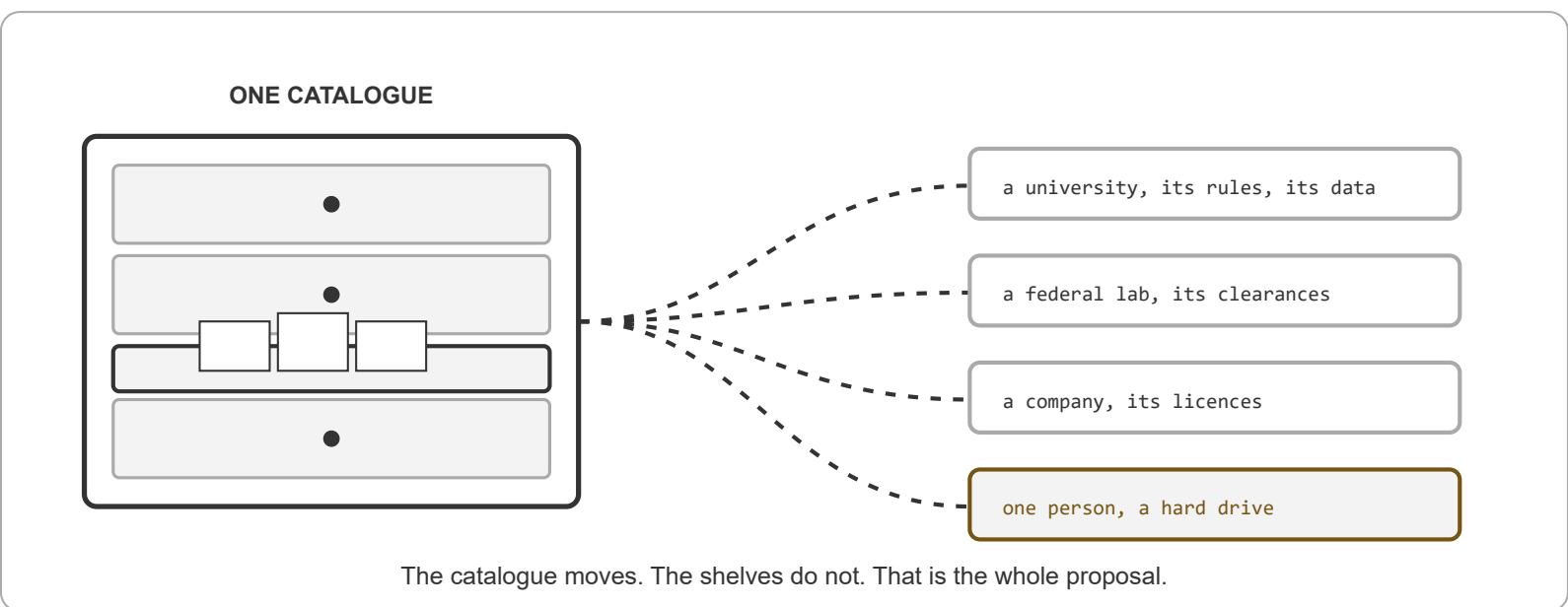
Sources: [Registered Reports](#)

Two ways to see the idea

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A federation, not a migration.



The catalogue moves. The shelves do not.

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Learning and continued support

When does continued support build capability, and when does it become dependency?

My example

An experiment may be blocked, completed with mixed results, or outside a tool’s valid range. One pass/fail label cannot explain all of that.

In the documented Mojo test, agreement passed; Float64 accuracy passed; Float32 accuracy failed. The overall protocol still failed.

documented

Sources: S02-I report

The Mojo result illustrates how to report a mixed technical outcome. It is not an SBIR award evaluation and does not establish a case for continued funding.

A practical next step

Record what another attempt changed: uncertainty resolved, capability gained, transition achieved, or a justified stop. Review this alongside commercial or mission progress.

Progress would mean: Evidence of learning and transition, with reasons for continuation, redirection, or stopping.

Who needs to be involved

Program evaluators, researchers, entrepreneurs, customers, and funding partners.

Research stage: Validation → Commercialization → Impact.

The SBIR discussion: hold both questions open

Cohen, Feldman, and Stern emphasize mission, capability, and stable support. Schönander asks how the wider ecosystem and opportunities for newcomers affect translation. Katz puts more weight on commercialization and graduation. The Murphy interview emphasizes results, learning, and transition.

Together, these perspectives invite a stronger review question: what did another award or attempt change? Documented learning can complement commercial or mission progress; it is not an automatic reason to keep funding the same work.

Evidence and technical detail

Foundations already exist. OSF supports research registration. Registered Reports evaluate research plans before results are known. The distinctive addition here is an interoperable record of execution, outcomes, reasons, and subsequent decisions, kept as separate dimensions rather than one pass/fail flag.

Dimension	Example values
Execution	Planned, not started, blocked, running, completed, aborted

Dimension	Example values
Validity	Valid, invalid, uncertain, outside applicable domain
Finding	Supports claim, contradicts claim, mixed, inconclusive, not evaluated
Criterion results	Backend parity passed; accuracy failed; timing unevaluated
Reason	Resource unavailable, toolchain problem, implementation defect, precision limit, insufficient data
Review	Unreviewed, machine-checked, human-reviewed, disputed
Subsequent decision	Continue, revise protocol, replicate, pause, retire
Record history	Current, corrected, superseded, withdrawn

Each classification keeps its source passage or artifact, applicable rule version, reviewer, and date. Distinguish when the experiment happened from when its record was reconstructed or published. Retrospective reconstruction is never labeled preregistration.

The Mojo campaign is the worked example: backend parity passed, Float64 accuracy passed, Float32 accuracy failed, timing unevaluated, overall protocol failed, decision: continue with Float64 and study Float32 separately. One label would have lost all of that.

Sources: [Center for Open Science, Registered Reports](#)

Agenda connection: [Resource Guide](#), p. 15; [SBIR participant handout](#), pp. 1–3. Suggested breakout: Track D · Funding and investment. Track placement and proposals are my contributions; the handouts provide the discussion context.

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WHAT THE RECORD SAYS

VS

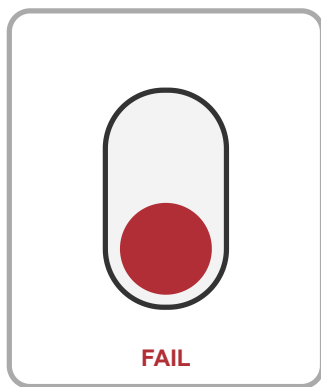
WHAT ACTUALLY HAPPENED

- both programs agreed, 24 of 24
- Float64 met the accuracy limit
- Float32 did not
- speed: never measured
- decision: ship Float64, study Float32

Same run. Five different facts.

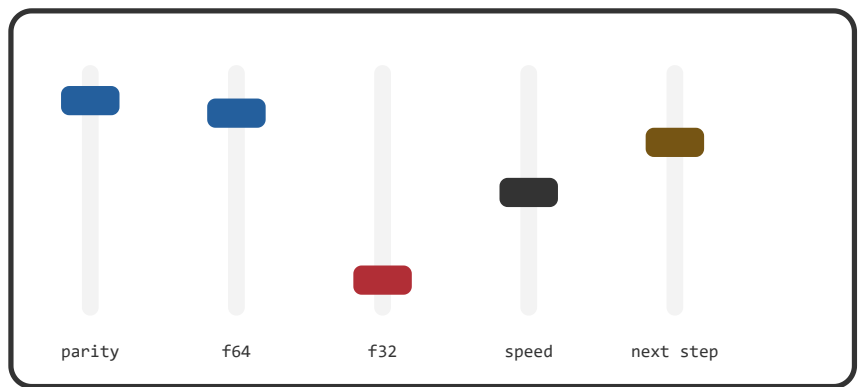
One label, or five facts. Same experiment.

one switch



VS

five independent settings, from one run



The run did not produce one verdict. It produced five, and they do not agree.

The run did not produce one verdict. It produced five.

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Evidence stewardship

What infrastructure makes a result accessible and understandable later?

My example

Large research artifacts need affordable storage, findable metadata, integrity checks, and links that last. Capacity alone does not make evidence understandable or reproducible.

A practical next step

Connect reports to retained artifacts, integrity checks, clear access terms, and a named custodian.

Progress would mean: An authorized reviewer can find, verify, and interpret the retained record.

Who needs to be involved

Researchers, repositories, storage providers, and data stewards.

Research stage: Validation → Reuse.

More context

- Independent researchers need clear guidance on eligibility, costs, retention, and preservation responsibilities.
- Pair storage allocations with curation and preservation help.

Evidence and technical detail

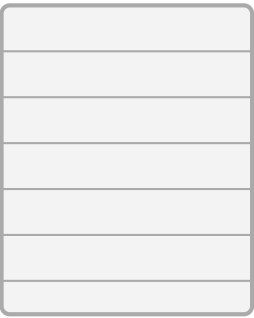
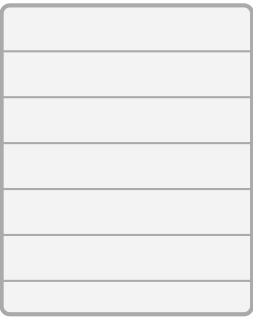
The supplied Mojo explainer identifies an externally retained raw record that this presentation has not independently inspected. The general stewardship need is clear: connect a report to identifiable artifacts, integrity checks, retention terms, and a usable access path.

A storage allocation becomes more useful when another researcher can locate and understand the evidence, subject to its access and licensing rules.

Agenda connection: [Resource Guide](#), p. 14. Suggested breakout: Track C · Innovation infrastructure. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



rent personal card
catalogue none
successor none

One key. One card. One person who remembers what is in there.

Capacity was never the hard part.

One key, one card, one person who remembers what is in there.

it shipped · it arrived · it is intact

Nobody can
clear it.



Nobody can
use it.

Bytes preserved perfectly, and stuck in port, because nobody funded the paperwork.

Bytes preserved perfectly, and stuck in port.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Exploratory research

How can early exploratory work be understandable without overstating readiness?

My example

This is exploratory computational research with explicit limits. Its value for the Roadmap is the practice, not the hypothesis: numerical checks, failed attempts, corrections, and open questions are all kept and shown.

Exploratory work can produce useful methods and infrastructure even when its motivating hypothesis stays unproven.

documented

Mathematics and code only; no demonstration of physical feasibility.

Sources: [Warp Field Notebook](#)

A practical next step

Use a bounded notebook to separate the question, assumptions, mathematical or code results, and unresolved tests.

Progress would mean: A reviewer can distinguish an exploratory result from a validated application.

Who needs to be involved

Researchers, technical reviewers, and possible collaborators.

Research stage: Discovery → Translation.

More context

- Present it as exploratory research with stated limitations.
- Show how numerical checks, failed attempts, corrections, and unresolved questions are retained.
- Highlight reusable tooling and validation methods produced along the way.
- Add an experiment ledger with dates, protocol versions, criterion-level outcomes, reasons, and evidence links.
- Distinguish contemporaneous records from later reconstruction.

Evidence and technical detail

Three exhibit records the logs already support:

1. An experiment blocked by an incorrect environment diagnosis, later corrected.
2. A port that matched its reference implementation but failed an independent accuracy requirement.
3. Correct numerical output with unresolved or insufficient performance evidence.

For each ledger row: experiment and attempt IDs, execution date, protocol version, code and environment identifiers, criterion-level outcomes, reason codes, evidence links, and corrections. Where the logs do not establish whether something ran, the ledger says *unknown*.

Neither backend agreement nor the stencil qualification demonstrates a realizable propulsion system. That sentence stays attached to the notebook wherever it is presented.

Sources: [Computational-Spacetime-Dynamics-Research](#) · [temporal-modulation-alcubierre](#)

Agenda connection: [Resource Guide](#), p. 3. Suggested breakout: Track A · Barriers to translation. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.

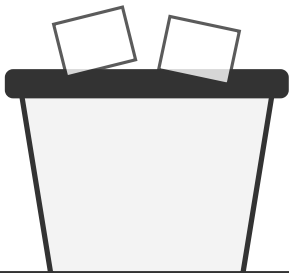
WHAT WORKED	WHAT DID NOT — KEPT ANYWAY
stencil ported, runs on gfx1103	Float32 missed accuracy
Float64 within frozen limits	TOOLCHAIN_ABSENT
byte-identical to HIP, 24 / 24	↳ wrong. Pixi env was there.
tooling reusable by other work	speed: never measured
	overall protocol: failed
	unused

The hypothesis is still unproven. The methods, the tooling and the corrections are not.

The eraser goes unused. That is the method.

A negative result is not waste. It is a narrowed question.

KEPT, ON THE SHELF



VS

blocked · wrong environment · corrected ↳ saved the next person two weeks
failed a limit · precision · f32 ↳ told us exactly where to stop
inconclusive · not enough data ↳ still narrows the question

USUAL PRACTICE

the result nobody sees again

each one still doing work

A negative result is not waste. It is a narrowed question.
These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Evidence exhibit

Can an unfamiliar participant understand what was tested and what remains unresolved?

My example

The proposed convening exhibit would let a visitor inspect one research record: what was tested, what happened, which evidence supports it, and what changed after review.

The participant workflow is in development. This presentation is not a live Observatory connection.

in development

A practical next step

Test a reviewed, sanitized research record as a small exhibit before claiming a ready participant workflow.

Progress would mean: Visitors can explain the test, result, evidence, and remaining uncertainty.

Who needs to be involved

Exhibit builders, research custodians, reviewers, and intended visitors.

Research stage: Validation → Communication.

Evidence and technical detail

The intended workflow distinguishes an authored replay, a live Engine result, and a proposed capability. Any live participant demonstration needs separate readiness evidence, including access controls and the approved corpus.

Semantic or graph search would aid discovery. The source record remains the basis of the claim. These are demonstration requirements, not a statement that every integration is ready.

Sources: [Observatory repository](#) · [Release record](#)

Agenda connection: [Resource Guide](#), p. 15. Suggested breakout: Track B · Ecosystem best practices. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.

EXPERIMENT S02-I

parity PASS

Float64 PASS

Float32 FAIL

speed UNEVALUATED

overall FAIL

● v1 • recorded

● v2 • corrected

● v3 • current

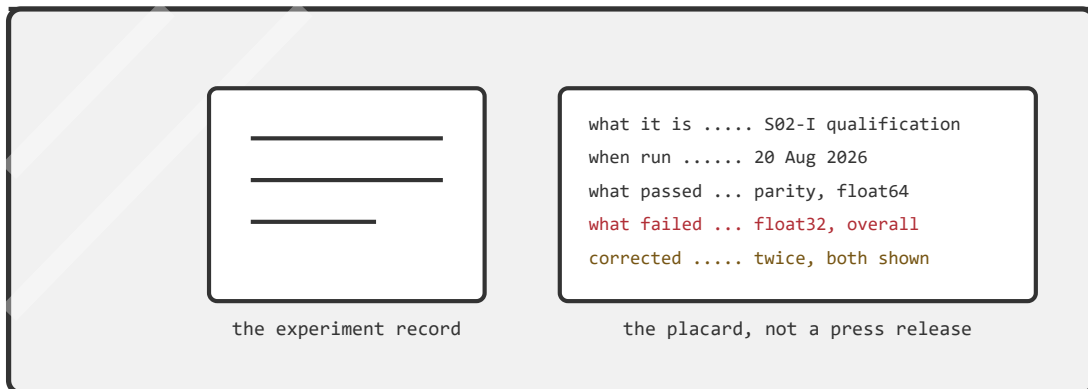
nothing was deleted to get here
every chip opens its own evidence

READ-ONLY
EXHIBIT

Under development. The exhibit will be labelled replay or live, never left ambiguous.

Every chip opens its own evidence. Nothing was deleted to get here.

You may look. You may not touch. The label tells the truth.



A read-only exhibit is not a limitation we apologise for. It is the exhibit's integrity.

status: in development • the exhibit link appears here once verified

You may look. You may not touch. The label tells the truth.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Community partnerships

How can research institutions build useful, lasting relationships with communities?

My example

Florida Armored Combat is a nonprofit whose members include veterans, parents, blue-collar and white-collar workers, and people from many backgrounds. We have a community and worthwhile questions about belonging, recreation, participation barriers, and retention. We do not have a research partner.

Question for the room. Where does a nonprofit go when it has a community and good questions, but no research partner?

Community observations suggest questions; they do not establish health benefits or causation.

A practical next step

Use Florida Armored Combat as a possible partner for shaping a research question together, before choosing a method or making an outcome claim.

Progress would mean: A jointly defined question, clear roles, an agreed method, and a useful return to the community.

Who needs to be involved

Community members, researchers, and an accountable institutional partner.

Research stage: Discovery → Validation.

More context

- Community observations can generate research questions. They do not by themselves establish effectiveness or causation.
- Researchers and community members should develop the questions together.
- Initial support should help establish partnerships, design studies, and arrange appropriate oversight.

Evidence and technical detail

A possible next step: ask a research institution for a community-engagement consultation. The University of Florida Clinical and Translational Science Institute provides a community engagement entry point to investigate. A sensible first project would ask why adults join, remain, or leave, and what benefits or barriers they perceive, including former participants so current members alone do not distort the picture. The institution determines review and consent requirements before any recruitment.

A short partnership brief would cover FAC's activities and locations, the membership diversity, questions members want answered, what FAC can contribute (relationships, facilities, organizational knowledge), and what it needs (study design, independent analysis, research oversight).

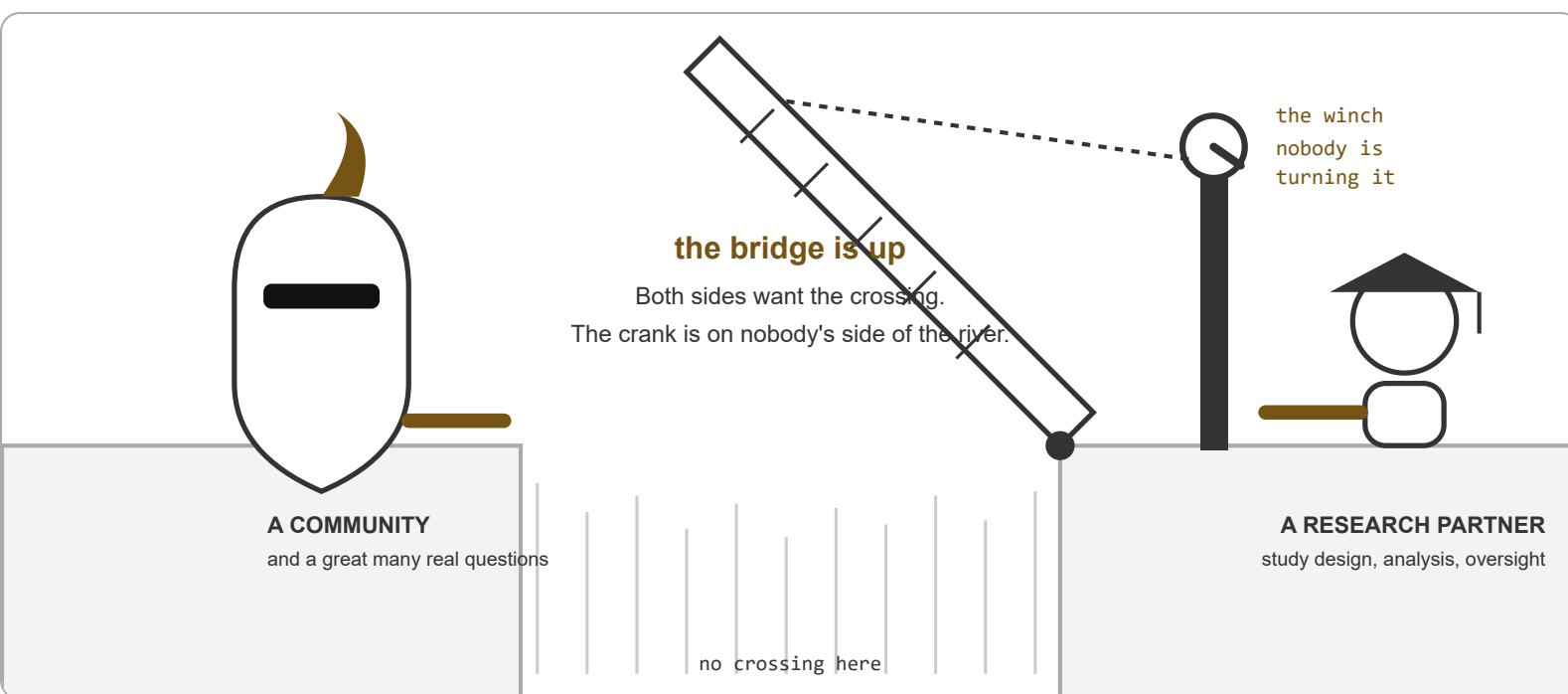
This is the portal use case in miniature. The request is not money first. It is a named person who can make the introduction and stay responsible for it.

Sources: [UF CTSI community entry point](#) · [Florida Armored Combat](#)

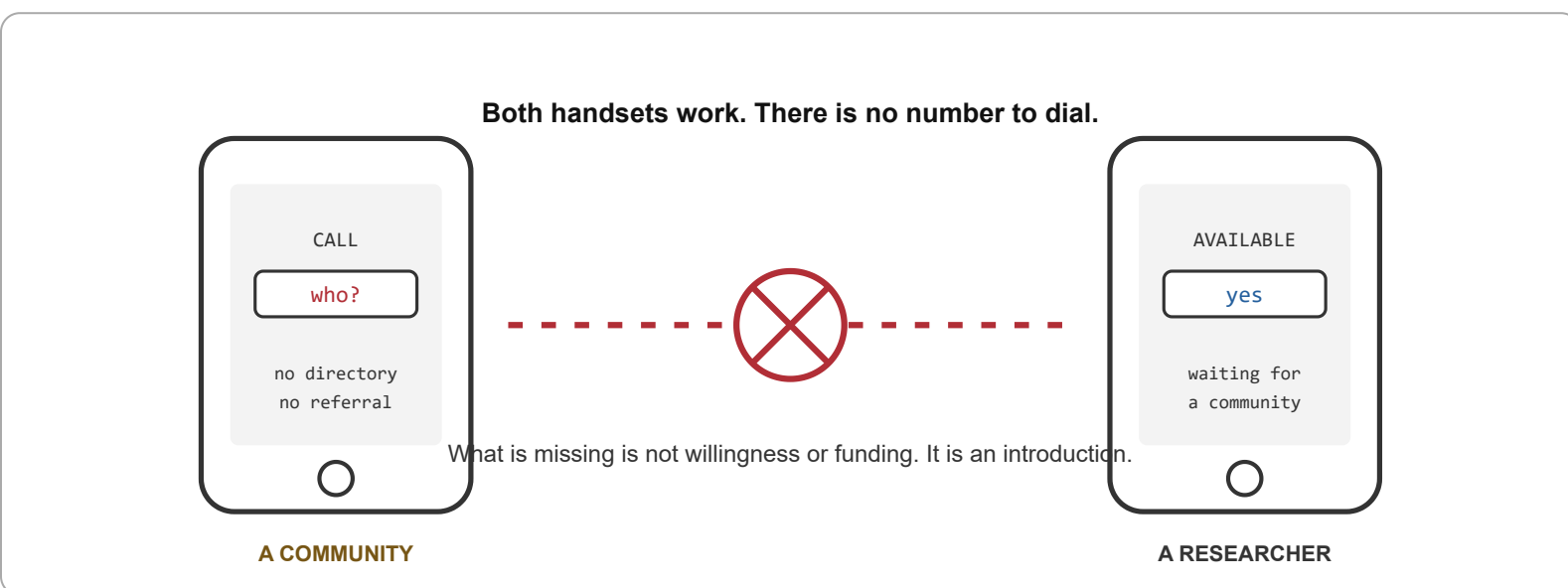
Agenda connection: [Resource Guide](#), p. 14. Suggested breakout: Track B · Ecosystem best practices. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



The crank is on nobody's side of the river.



Both handsets work. There is no number to dial.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Technical assistance

What expertise is missing between a research result and a usable tool?

My example

Some projects need hands-on engineering help to become usable and maintainable: research software engineering, validation, deployment, security, data stewardship, documentation. Define that service and its users before deciding whether it needs a new institution.

Question for the room. Who helps promising work cross the distance between a working prototype and something others can rely on?

A practical next step

Pilot a small engineering and validation service with clear intake, scope, ownership, and handoff.

Progress would mean: Useful handoffs, reproducible work, external adoption, and a maintenance plan.

Who needs to be involved

Research software engineers, domain specialists, maintainers, and regional partners.

Research stage: Translation → Validation → Adoption.

More context

- NSF already has a technology-and-translation directorate, TIP, whose role includes partnerships with industry, nonprofits, civil society, and communities of practice.
- The proposed addition is accessible engineering assistance: research software engineering and maintenance, reproducibility and independent validation, secure deployment, data stewardship, help moving prototypes into sustained use.
- Test it as a funded network of technical assistance centers coordinated through existing structures. A new federal institution is a much larger claim that has not been shown necessary.

Evidence and technical detail

How the three proposals fit together: the portal is the entry point, the technical network provides the assistance, and the research library preserves the evidence and learning. An Observatory exhibit could illustrate the research-record part of the proposal.

Service	Who needs it	Existing analogue
Research software engineering and maintenance	Independent maintainers, small labs	University research software engineering groups; access varies
Reproducibility and independent validation	Anyone making a technical claim	Registered Reports, replication grants
Secure deployment and infrastructure	Nonprofits, community projects	Cloud credits (capacity without stewardship)

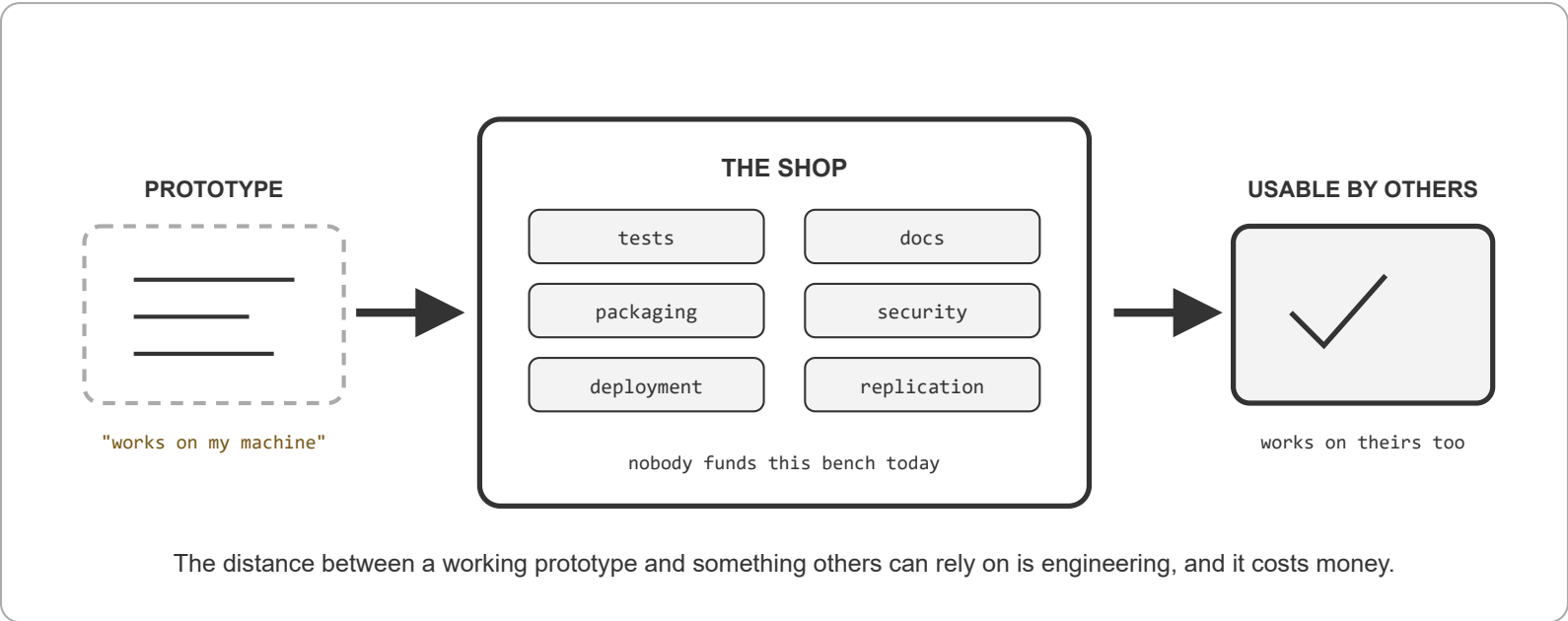
Service	Who needs it	Existing analogue
Data stewardship and record interoperability	All of the above	Domain repositories, OSF

Sources: [NSF TIP](#)

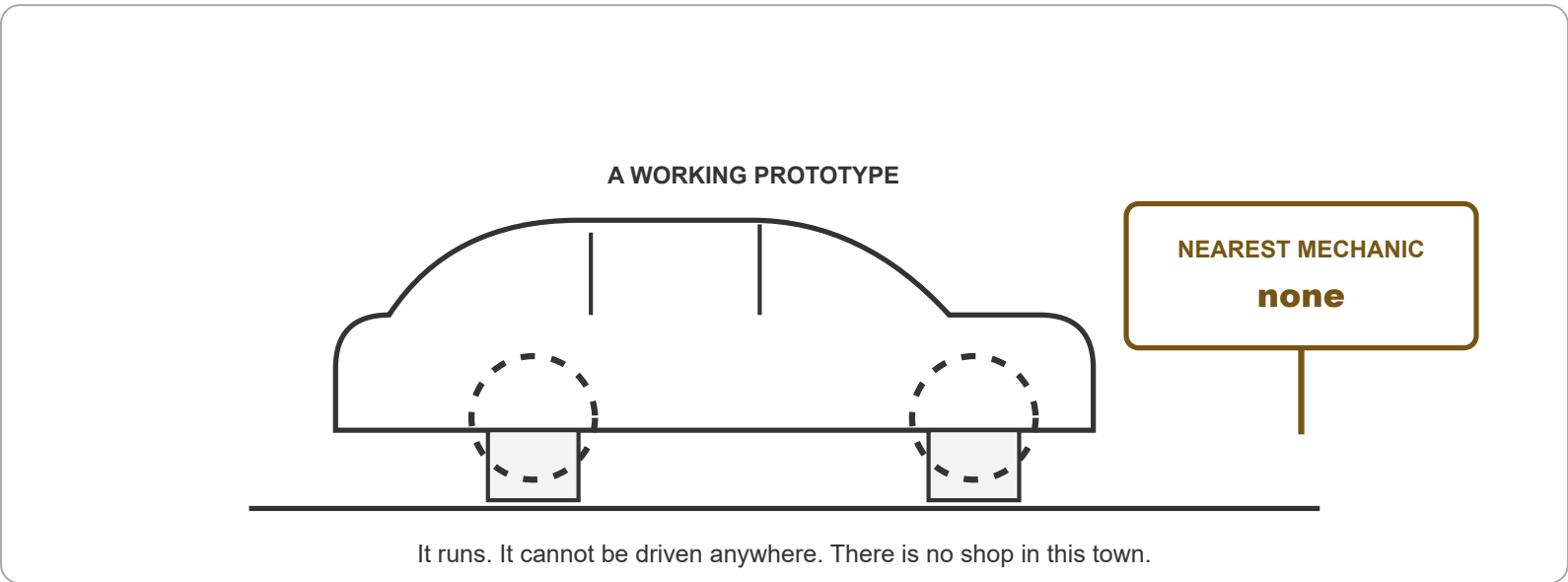
Agenda connection: [Resource Guide](#), p. 2. Suggested breakout: Track C · Innovation infrastructure. Track placement and proposals are my contributions; the handouts provide the discussion context.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Works on my machine, into works on theirs. That gap is engineering, and it costs money.



It runs. It cannot be driven anywhere.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Evidence you can open

These six repositories are the starting points identified in the supplied materials. Specific result claims should be checked against their dated reports. Repository access and current readiness may vary.

- [Computational-Spacetime-Dynamics-Research](#): the research campaign records
- [KosMojAMD](#): the Mojo on AMD modernization case study
- [gkos-standard](#): the evidence-governance pre-standard
- [GKOS-Engine](#): the TypeScript reference engine
- [GKOS-Engine-Rust](#): the Rust implementation effort
- [GKOS-Observatory](#): the demonstration

What useful evidence records should preserve:

- Exact experiment and source identifiers.
- Failed, blocked, superseded, and inconclusive outcomes.
- Original timestamps, with execution date kept separate from publication date.
- Hardware, software, and environment boundaries.
- Links from summaries to the underlying reports, manifests, logs, and artifacts.
- Corrections as visible successor records, never silent rewrites.
- Explicit separation of backend agreement, scientific accuracy, performance, and production readiness.

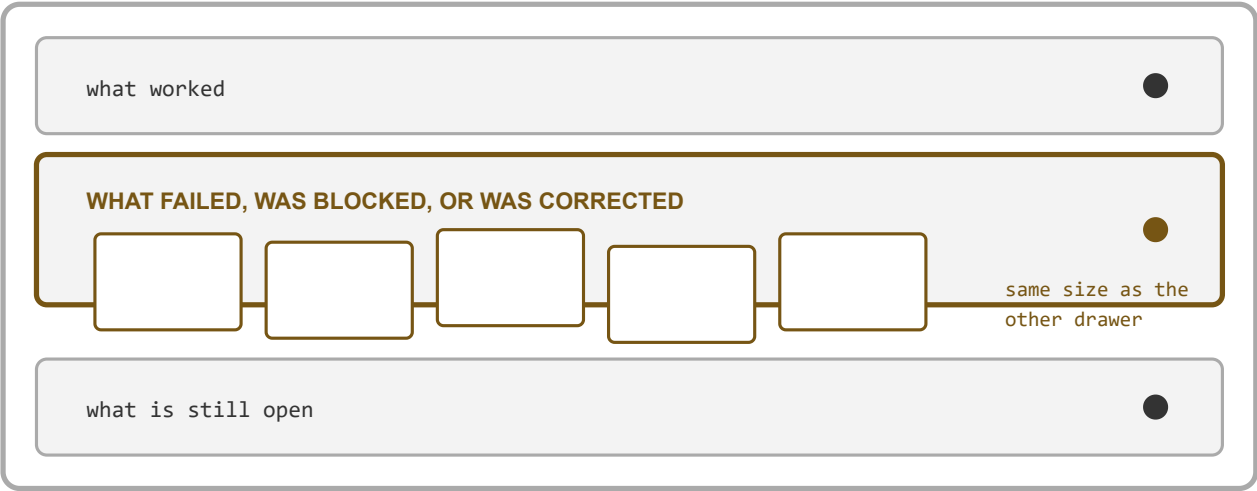
Related work, identified in the supplied materials, with current status requiring review: LLM_Engine_Rust-Over-GGML (archived comparison), LLM_Engine_Mojo, LLM_Engine_MojoRust (gated rebuild), GKOS-Engine-Lite, gkos-hindsight-governance, model-capability-manifest-standard, Kosmos-Oden, Kosmos-Oden-Lite, Kosmos_Research_Studio_Lite, Marshal-008-OS, theMarshal-Core-Rust, Hypatia, temporal-modulation-alcubierre, Paper01.

The three Rust/Mojo engine repositories overlap historically. Their roles and evidence boundaries need to be reconciled so a visitor can tell immediately what was attempted, what was measured, what was learned, and which repository now carries the authoritative evidence.

Before any private repository goes public it gets a disclosure review: credentials, personal data, internal network details, large artifacts, licensing, and any claim that exceeds the retained evidence.

Two ways to see the idea

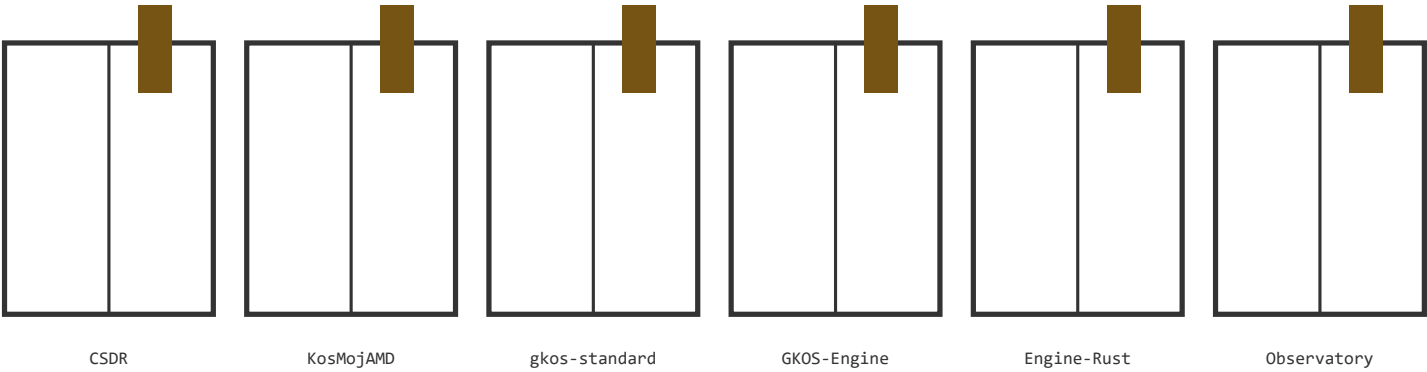
Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Six public repositories. The failure drawer is the one worth opening first.

The failure drawer is the one worth opening first.

Six books. Every one has the receipts bookmarked.



Open any of them at the amber marker. That is where the failures are.

Open any of them at the amber marker. That is where the failures are.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Three pilot proposals

- **Point to an existing pathway.** If a program already serves independent contributors well, say which one and how someone gets in.
- **Offer technical review.** Pick one repository and check whether its claims match its evidence.
- **Connect a research partner.** FAC has a community and questions. It needs a study-design collaborator.
- **Help shape a pilot.** Portal, technical assistance network, or federated research records.

For the Roadmap facilitators, the asks map onto the seven breakout points and Tracks A through D. See the mid-level and deep views of each talking point for the track tags.

Contact: Shaun "Oden" Marshall, shaunmarshall.com/cv. Florida Armored Combat: flarmored.org.

Pilot	Accountable partner to identify	Measure
Supported referrals	Receiving program and named navigator	Time to a useful connection; referrals completed
Technical assistance	Engineering or validation team	External adoption; reproducible results; maintenance continuity
Reusable research records	Source custodians and reviewers	Review effort; missing evidence; reusable learning

Start with a bounded pilot, an accountable partner, and a measure of success. Preserve source records and limitations so later reviewers can assess what the pilot established.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



NAME A PATHWAY
one that already works



CHECK MY WORK
pick any repository



INTRODUCE US
a research partner for FAC

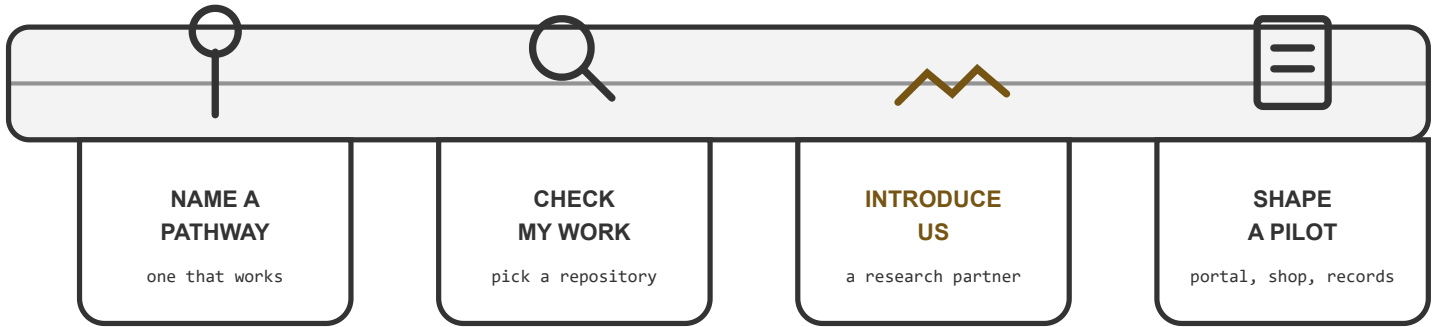


SHAPE A PILOT
portal, shop, or records

Four things you can do at this table that cost nothing and change something.

Four things you can do at this table.

Take whichever one fits your hand.



None of these costs a budget line. All of them cost a conversation.

Take whichever one fits your hand.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

A possible product

What would turn a research component into something people can use and support?

My example

A supported numerical kernel library is a product hypothesis to test with researchers and engineering teams. The first questions are which workload matters, which hardware they use, and what evidence would make the tool useful.

Proposed product. S02-I qualified specific criteria for one tested stencil; Float32 accuracy and the overall protocol failed. No library-wide accuracy, speed, cost, or physical-feasibility claim follows from it.

Concept illustration, not live computation or a validated physical simulation. AMD and Mojo are product names, not endorsements.

A practical next step

Run customer discovery alongside a bounded qualification and packaging pilot.

Progress would mean: Evidence of user need, reproducible qualification, and a clear support plan.

Who needs to be involved

Prospective users, numerical specialists, maintainers, and translation partners.

Research stage: Validation → Commercialization.

More context

Start with customer discovery and a bounded technical pilot. A prospective user should help choose the workload and acceptance criteria. Qualification on additional hardware, reproducible packaging, documentation, and a support plan are separate milestones.

Evidence and technical detail

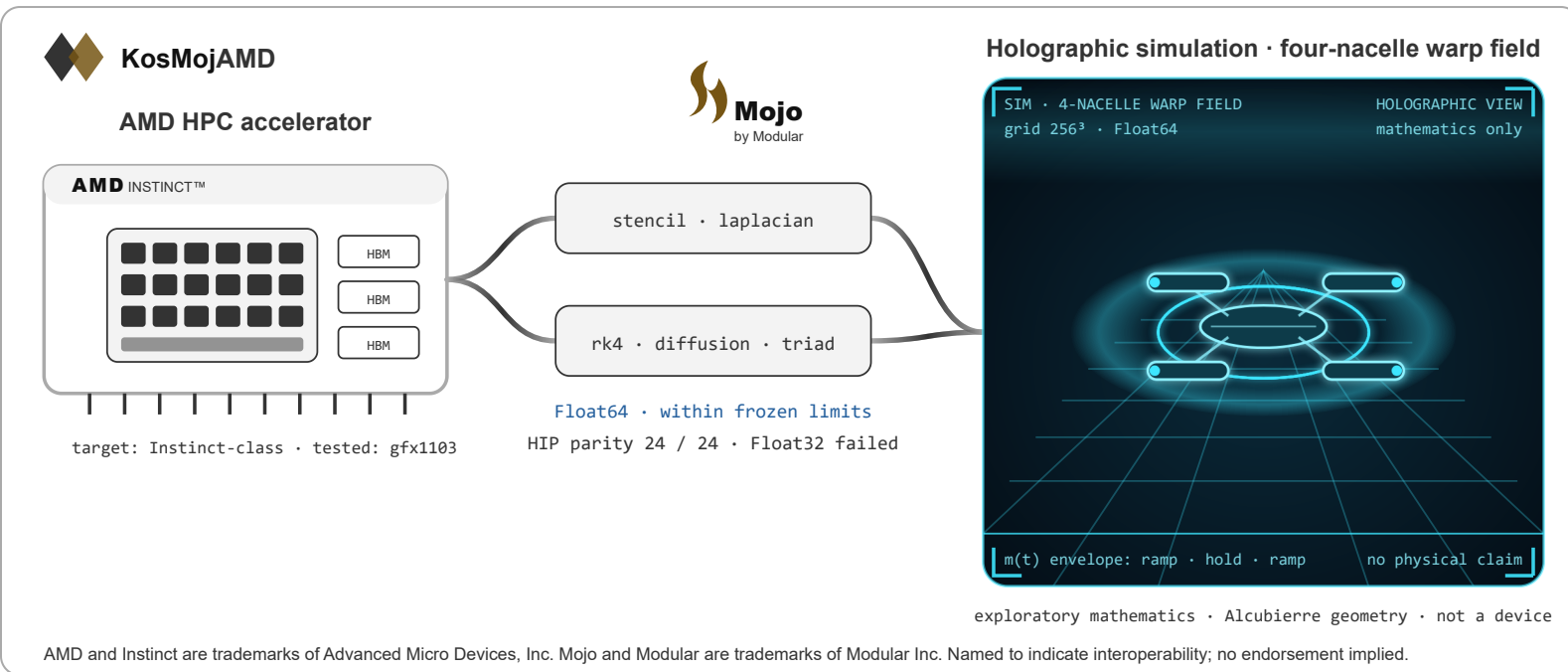
Evidence needed before a product claim

- Criterion-level numerical results for each supported workload and precision.
- Testing on the intended hardware and software versions.
- A benchmark with a defined baseline, environment, and repeatable method.
- User interviews and a pilot that tests usefulness, not just technical novelty.
- Clear licensing, maintenance ownership, and a sustainable support arrangement.

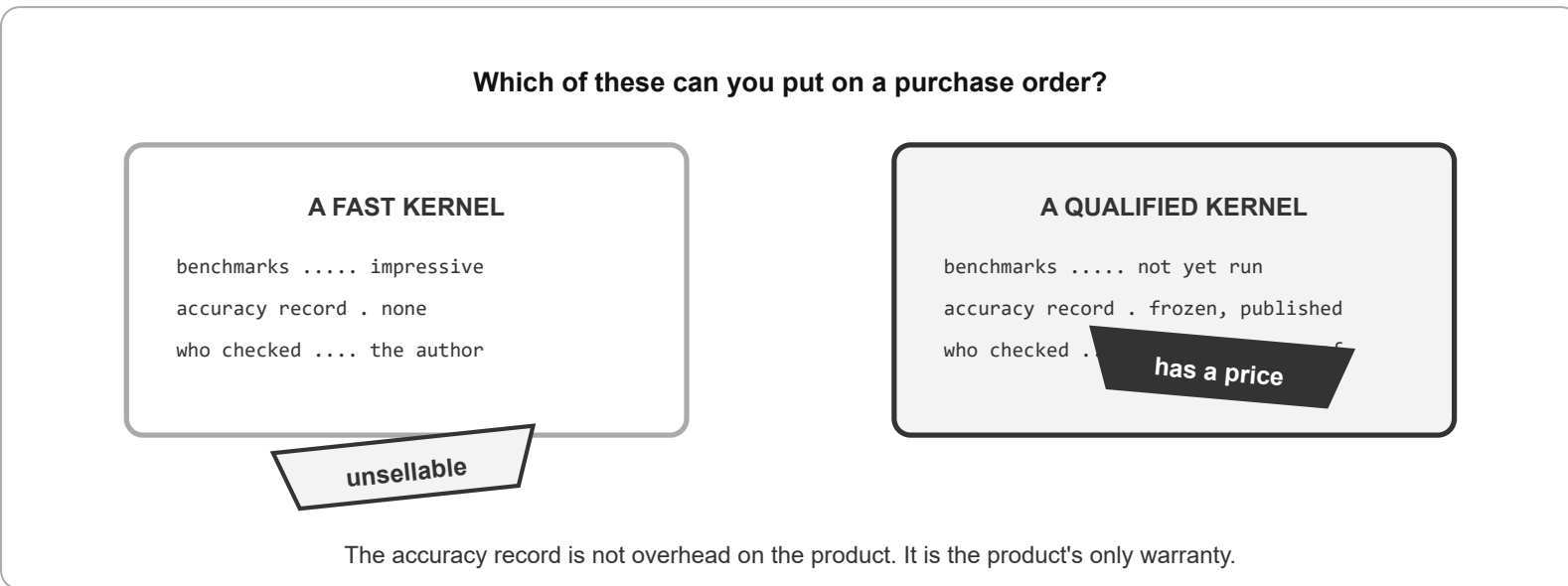
No price advantage, vendor portability, or customer demand has been established by the S02-I report. Exploratory spacetime work can motivate a workload without establishing physical feasibility.

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



AMD HPC accelerator running Mojo kernels to drive a holographic four-nacelle warp-field simulation.



The accuracy record is not overhead. It is the only warranty.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Three changes in five years

The Resource Guide asks each participant one question: if the United States could make only three major changes in the next five years to accelerate discovery and commercialization, what should they be? Here are mine.

Access

A supported front door, with a named person responsible for the first handoff, for people and groups with no institution behind them.

Capacity

Fund the engineering, maintenance, and independent validation that make a discovery usable, as translation work in its own right.

Learning

Shared, interoperable records of what was tried, what happened, why it stopped, and what should happen next.

Change	Barrier it removes (Guide question 1)	Intervention (Guide question 2)	Journey stage (Guide question 3)
Access	Eligibility without a usable path in; no introductions, no grants staff	Navigation and referral pilot with accountable handoff	Discovery to Translation
Capacity	Prototype cannot become something others rely on; maintenance unfunded	Technical assistance network; stewardship funded as translation	Translation to Validation to Scale
Learning	Results reduced to pass/fail; failures lost; repeated review	Federated research records with a small common schema	Validation, and evaluating repeated support

The Guide frames the convening as a move from observation to barrier, root cause, intervention, and action. Each of the three changes is written that way on this site: the barrier is stated at the surface, the root cause and the intervention at mid-level, and the evidence and open questions in the deep dive.

They also answer the Guide's question 7, access to innovation: who has access to the ecosystem and who does not. Independent contributors, small nonprofits, and community organizations are the people these changes are for.

Two ways to see the idea

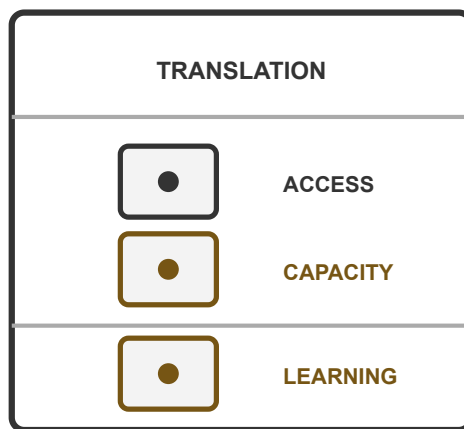
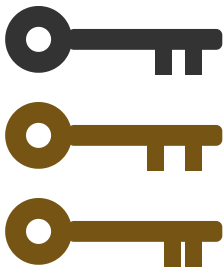
Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Three changes. Five years. Each one is a pilot somebody could fund next quarter.

The whole position, in three circles.

three keys



three locks

Turn one and the door still does not open. This is why the three arrive together.

Turn one and the door still does not open.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Where this fits in the convening agenda

Use the agenda as a conversation map: barriers, commercialization, capitalization, partnerships, and the four Roadmap tracks. These are suggested connections for my examples.

Agenda item	What I bring
Day 1 dialogue: the most consequential gaps	The gap between eligibility and entry for independent contributors; the gap between prototype and maintained, validated software.
Session 1: Commercialization	A proposed supported kernel library: technical qualification and customer discovery are separate milestones.
Session 2: Capitalization	Small, staged awards for qualification, replication, and packaging; evaluate repeated support by resolved uncertainty.
Session 3: Scaling through partnerships	Community-to-research matching (FAC case); technical assistance network through existing structures.
Track A: Barriers	Investigate barriers faced by people without institutional support.
Track B: Best practices	Preregistered gates, retained failures, visible corrections; open-source stewardship.
Track C: Infrastructure for the under-resourced	Portal with accountable handoff; storage with stewardship; assistance network.
Track D: Funding pipeline	Records that let funders see progress beyond pass/fail; staged support tied to explicit qualification milestones.

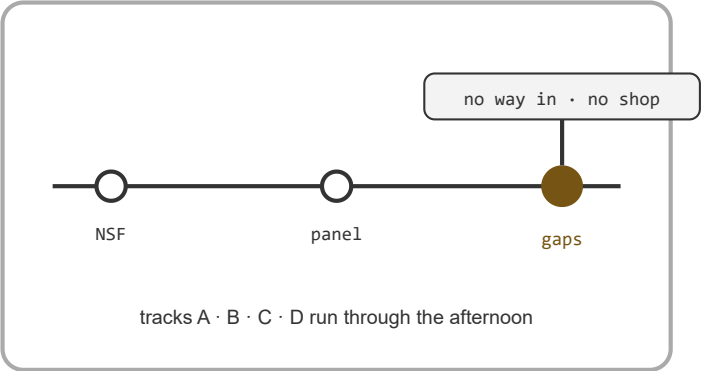
Seven breakout points, answered once for the whole position:

Point	Answer
1. Challenge	Capable independent contributors and community organizations cannot convert eligibility into entry, and prototypes cannot become maintained, validated tools.
2. Root cause	Investigate where missing staff, infrastructure, introductions, or maintenance support slows progress. The cause will vary by program and participant.
3. Existing models	SBIR/STTR, I-Corps, TIP Translation to Practice, OSF and Registered Reports, open-source foundations.
4. Gaps	Look for gaps in first handoffs, stewardship support, and interoperable evidence records.
5. Recommendations	Access pilot, capacity network, learning records: three bounded pilots.
6. Stakeholders	NSF TIP and SBIR/STTR, EPSCoR institutions, university CTSIs, open-source foundations, community nonprofits, NIST tech transfer.
7. Measures	Completed connections, progress toward validation, sustained reuse, resolved uncertainty. Not visits or award counts.

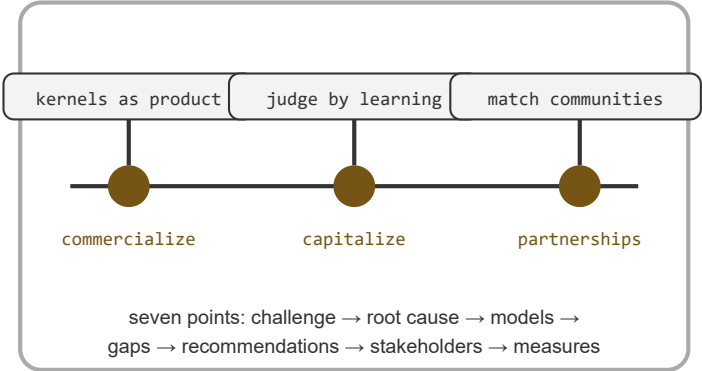
Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.

DAY 1 · BUILDING THE ROADMAP



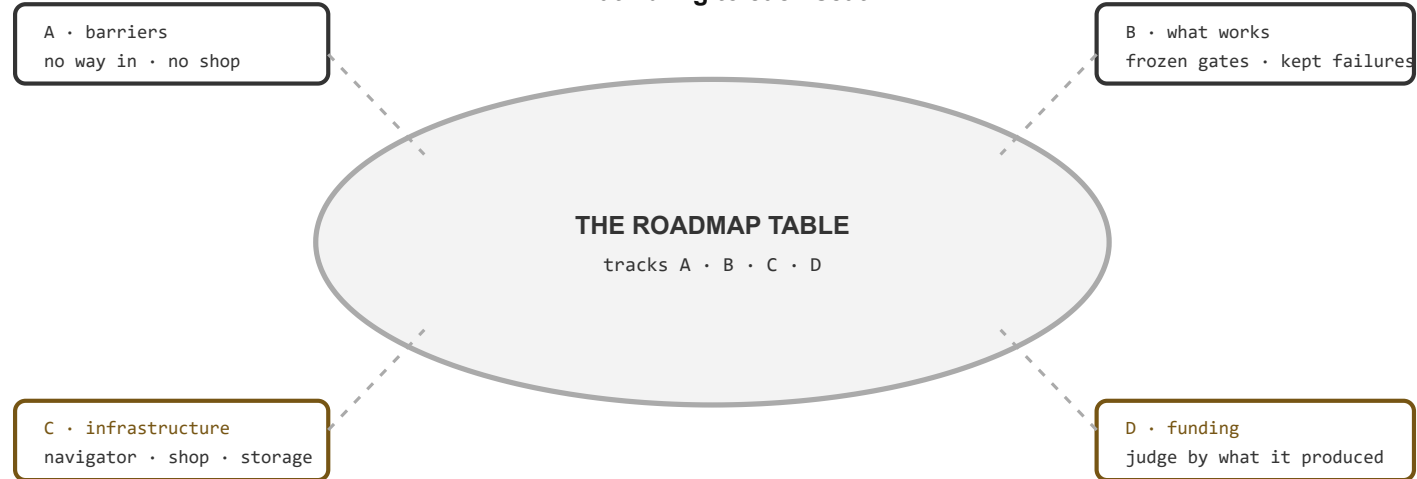
DAY 2 · DEVELOPING IT



Every topic on this site is tagged so it can be dropped into the right conversation.

Tagged so it drops into the right conversation.

What I bring to each seat



Every topic on this site is already tagged to one of these four.

Every topic is already tagged to one of these four.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

What the SBIR readings mean for people like me

The supplied SBIR handout presents competing views on stable support, commercialization, graduation, and opportunities for newcomers. A useful question across those views is: what did continued support change?

- Cohen, Feldman and Stern: SBIR is mission infrastructure; success looks different across agencies; use many measures and better integrated data.
- Schönander: reauthorization is not enough; open the door to first-time applicants; assess commercialization readiness explicitly; try lotteries and golden tickets for ideas consensus review misses.
- Katz: seed funding should graduate firms; weight commercial viability heavily; provide procurement pathways.
- The Murphy interview: ask what past awards produced; share past-performance and technical learning across agencies.

Common thread: reward demonstrable progress and learning. That needs records richer than pass or fail.

These are summaries of discussion positions in the handout, not current eligibility rules or funding decisions.

Where an independent contributor sits in that debate:

Reading's concern	What this position adds
New entrants are discouraged	A navigation service with an accountable handoff lowers the first step for individuals and informal teams.
Commercialization readiness should be assessed with evidence	Criterion-level records (parity, accuracy, performance, production readiness kept separate) are that evidence for technical products.
Past performance should capture learning, not award counts	Dispositions beyond pass/fail: blocked, inconclusive, failed-a-requirement, corrected. Publish them.
Fix the transition system, not only the award	Technical assistance network for validation, packaging, deployment, and documentation.

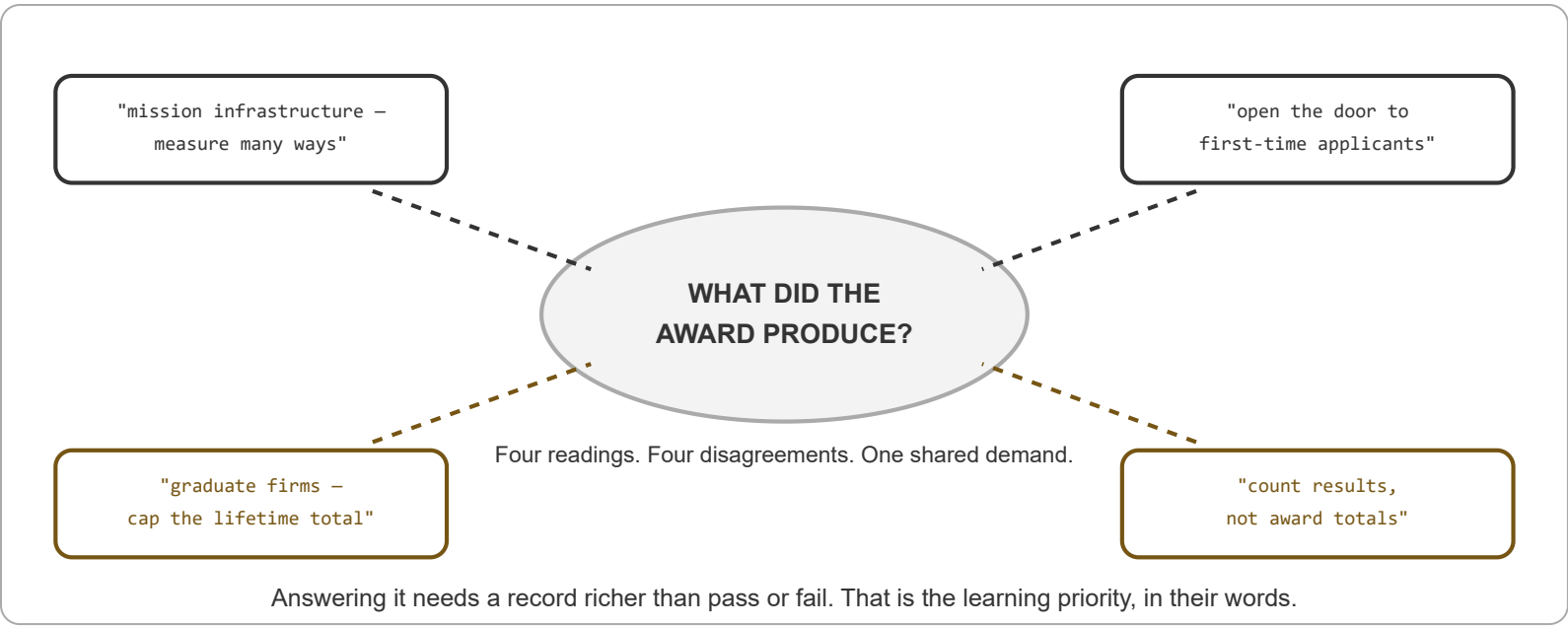
Sources: [Participant handout bibliography: Cohen, Feldman, Stern \(2026\)](#) · [Schönander \(2026\)](#)

Learning can complement commercial or mission progress; it does not automatically justify another award. The Mojo example is a technical reporting case, not an SBIR award evaluation.

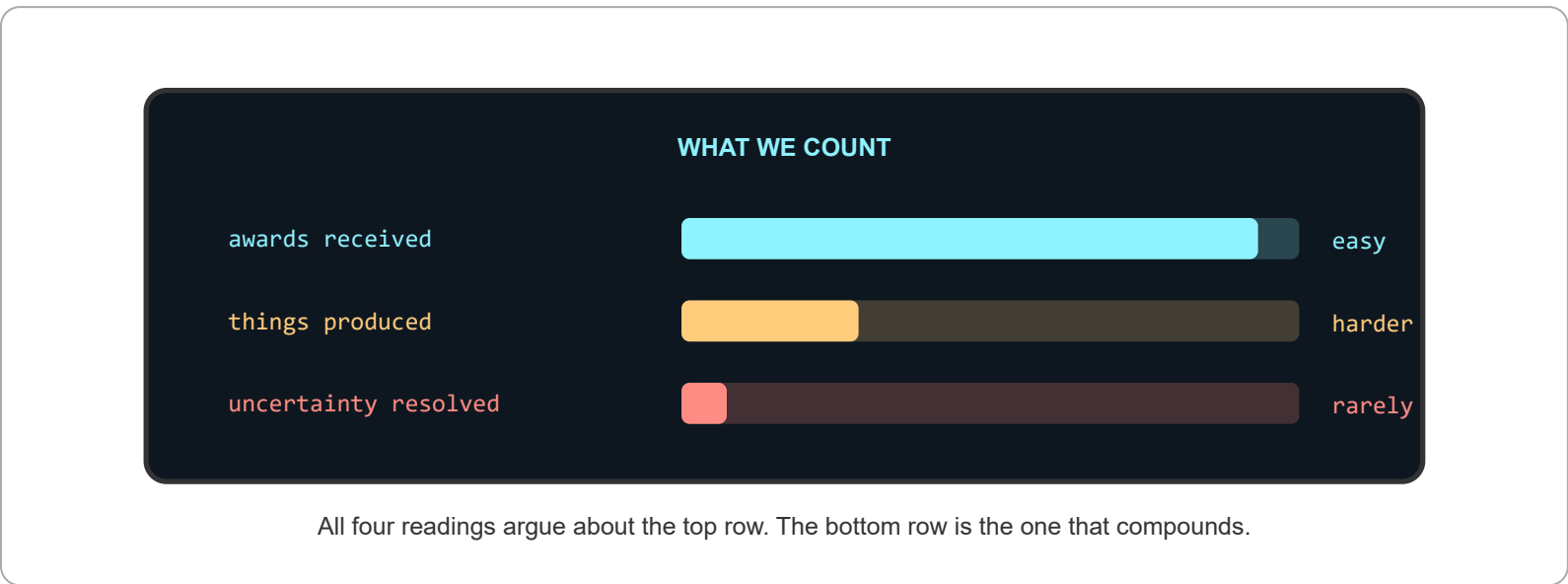
[Participant handout, pp. 1–3](#)

Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



Four readings. Four disagreements. One shared demand.



All four readings argue about the top row. The bottom row is the one that compounds.

All four readings argue about the top row. The bottom row compounds.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.

Where the journey slows down

The guide follows Discovery → Translation → Validation → Commercialization → Scale → Impact. My examples raise two practical questions: how does someone without institutional support reach translation, and how do they obtain the engineering needed for validation?

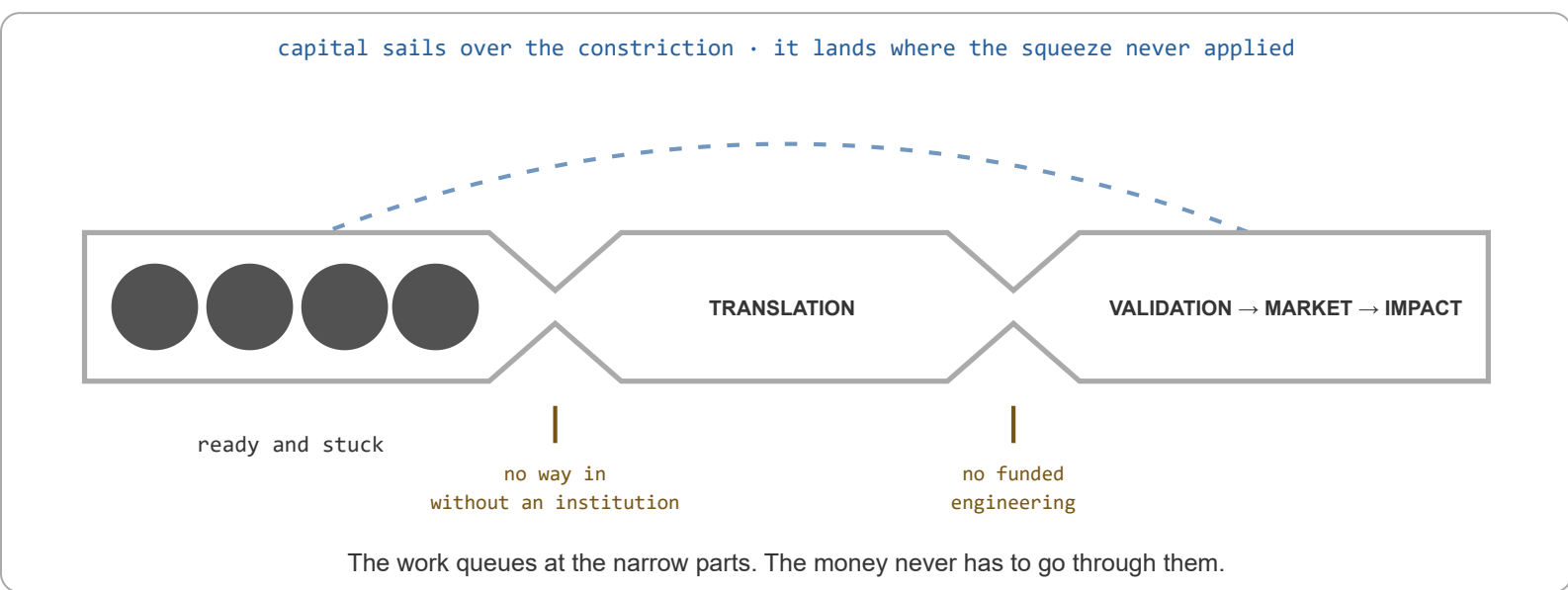
Stage	What slows it for independents	Which change helps
Discovery to Translation	No entry path, no introductions	Access
Translation to Validation	No funded engineering, no independent replication, no second hardware	Capacity
Validation to Commercialization	Evidence not legible to funders or buyers	Learning
Scale and Impact	Maintenance unfunded; dependencies rot	Capacity

AI may reduce effort in some tasks while creating more work for review in others. The guide's AI discussion asks where human judgment remains essential. The Mojo example supports the narrower practice of checking each technical criterion; it is not an evaluation of the AI studies summarized in the guide.

Resource Guide, p. 13

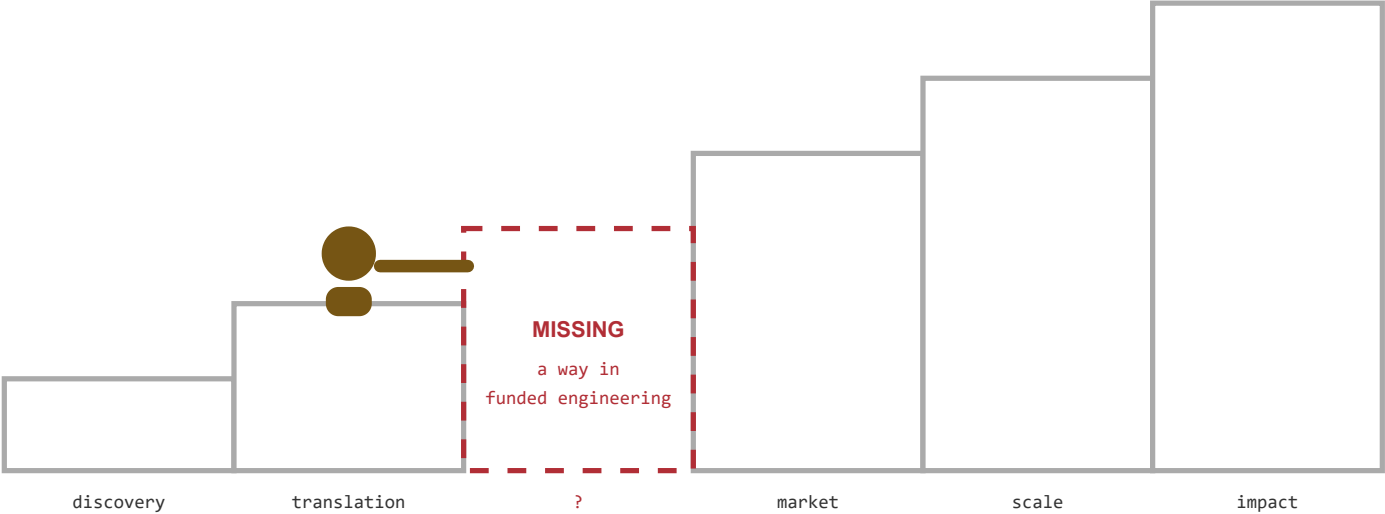
Two ways to see the idea

Choose the illustrations that help you. Hiding or swapping an image keeps every option in the [image appendix](#). Choices are saved on this browser.



The work queues at the narrow parts. The money never has to go through them.

The stair is not steep. A step is absent.



The stair is not steep. A step is absent.

These are explanatory illustrations, not additional experimental evidence. Concept drawings do not establish product readiness or physical feasibility.